
Chapter 8

Feedback

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Outline

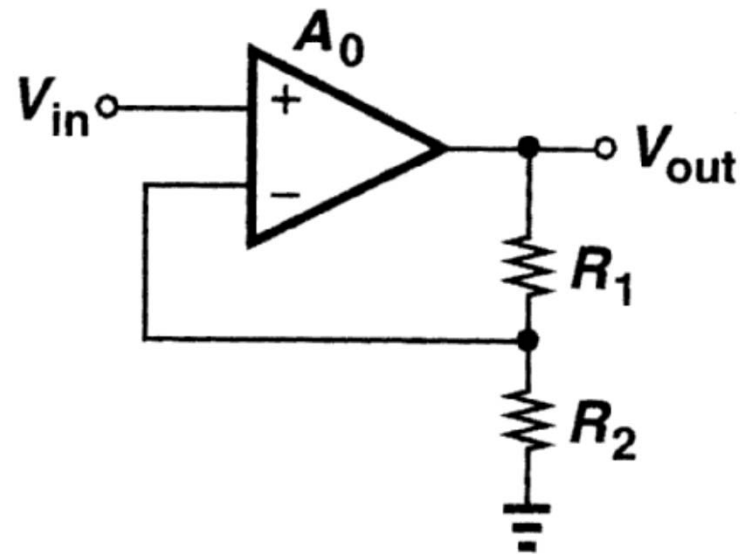
- **General consideration**
 - **Abstract model**
 - Properties of feedback circuits
- Feedback topologies
- Effect of feedback on noise
- Feedback analysis difficulties
- Two-port network and loading effect

Applications of feedback

- The idea of feedback was presented by Harold Black in 1927
 - On a ferry, a sudden idea to solve the nonlinearity in RF amplifiers
- The most important technique in analog design
- *Basic idea:*
 - A feedforward amplifier can provide high gain but with poor accuracy
 - Use negative feedback (often with passive devices) to obtain high precision
- *Positive feedback can be used to build oscillator*
 - *Will be discussed in chapter 14*



An example of feedback network

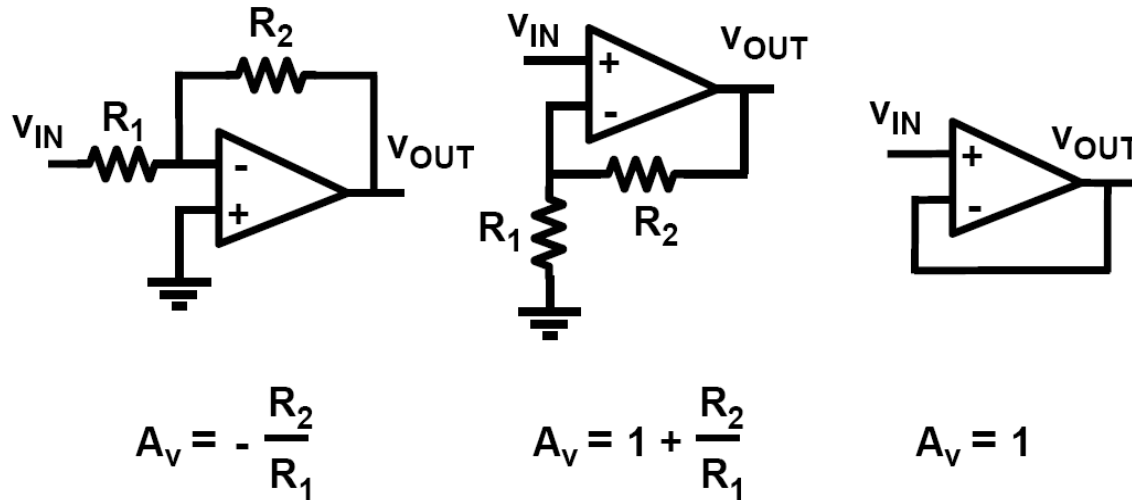


- If the amplifier has a large enough gain A_0

$$V_{in} = V_{out} \left(\frac{R_2}{R_1 + R_2} \right) \longrightarrow V_{out} = \left(1 + \frac{R_1}{R_2} \right) V_{in}$$



Negative-Feedback Operation



关键假设

虚断、虚短和虚地

虚断：运放输入阻抗为 $\infty \rightarrow i_{in}=0$

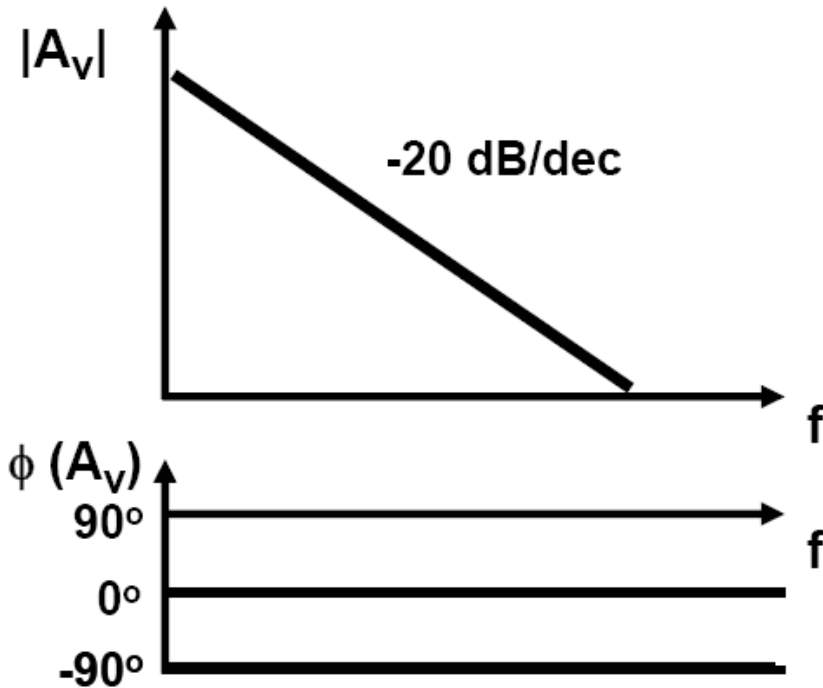
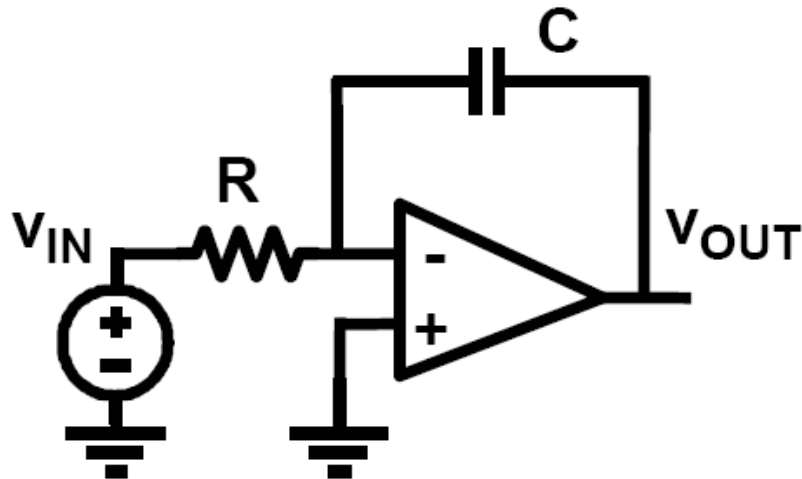
虚短：运放增益为 ∞ ，对于单位缓冲器有：

$$v_{out} = A_v(v_{in} - v_{out}) \Rightarrow v_{out} = \frac{A_v}{1 + A_v} v_{in}$$

——→ $A_v = \infty \Rightarrow v_{out} = v_{in} \Rightarrow v_+ = v_-$

虚地：运放一端接地，由虚短可得另一端电压必然为0

Integrator



$$\frac{V_{\text{out}}}{V_{\text{in}}} = \frac{1}{j\omega / \omega_p}$$

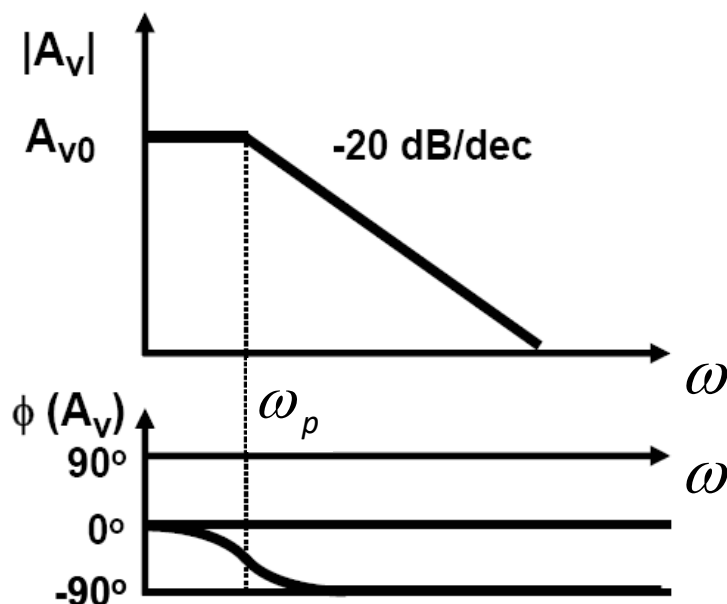
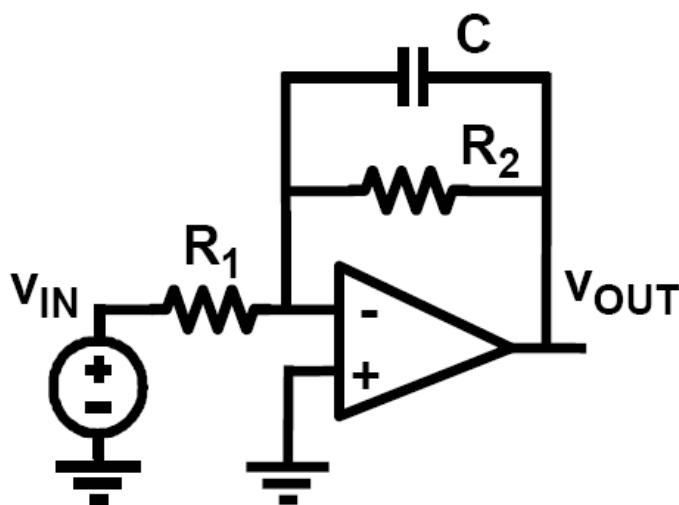
$$\omega_p = \frac{1}{RC}$$



Low-pass filter

- 有源RC滤波器

- 片上滤波器的一种重要实现形式



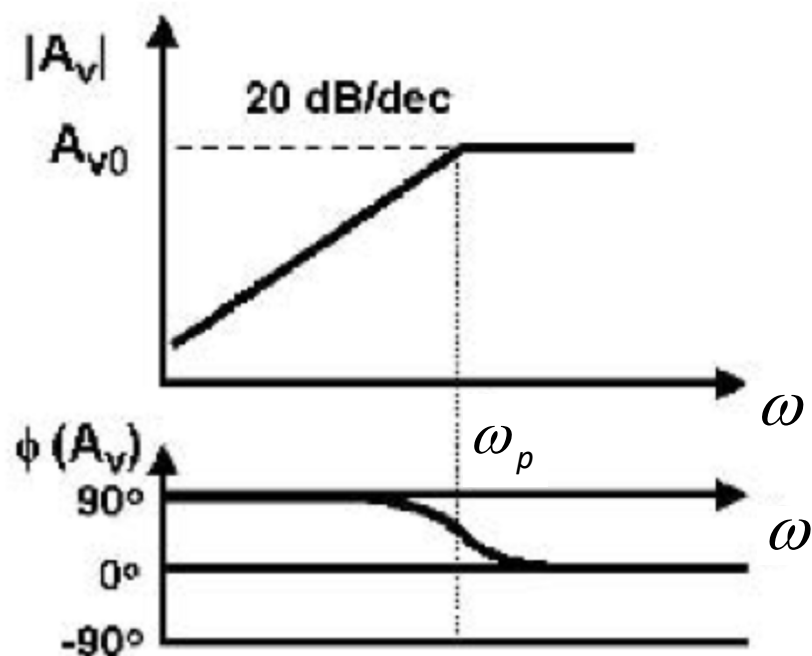
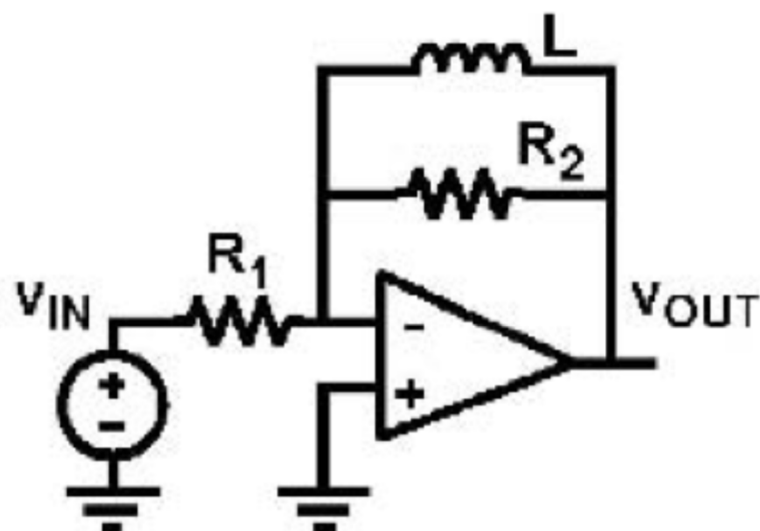
$$A_{v0} = -\frac{R_2}{R_1}$$

$$A_v = A_{v0} \frac{1}{1 + j\omega / \omega_p}$$

$$\omega_p = \frac{1}{R_2 C}$$



High-pass filter



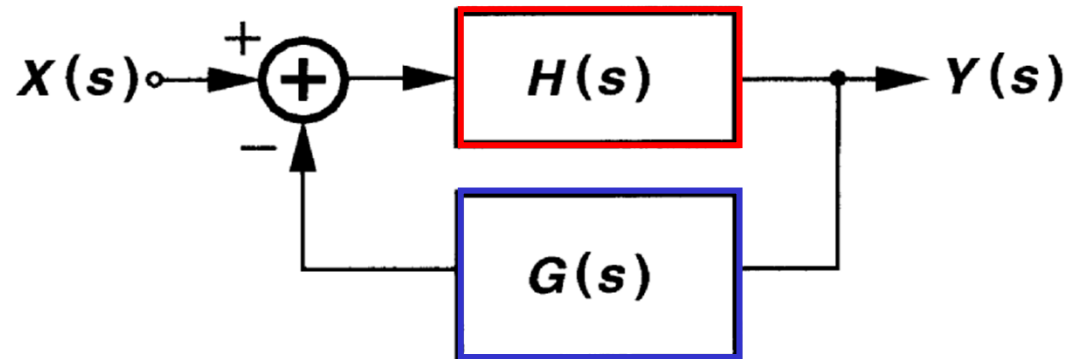
$$A_{v0} = -\frac{R_2}{R_1}$$

$$A_v = A_{v0} \frac{j\omega / \omega_p}{1 + j\omega / \omega_p}$$

$$\omega_p = \frac{R_2}{L}$$



Abstract model of feedback



- $H(s)$: feedforward network
 - An amplifier
- $G(s)$: feedback network
 - Frequency independent quantity
 - Usually defined as: β

- Key definitions:

- Close loop transfer function:

$$Y(s)/X(s)$$

- Open-loop transfer function:

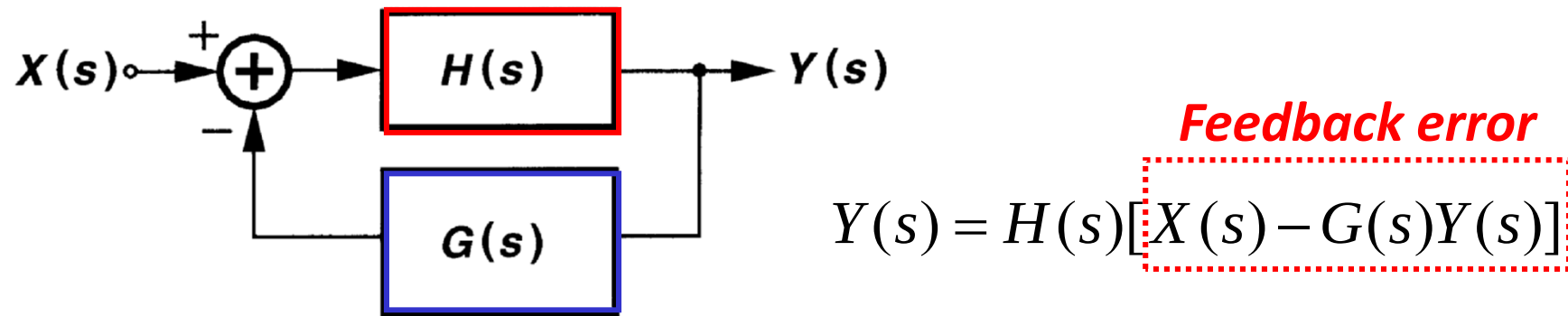
$$H(s)$$

- Loop transfer function(gain):

$$H(s)G(s)$$



4 basic elements in feedback networks



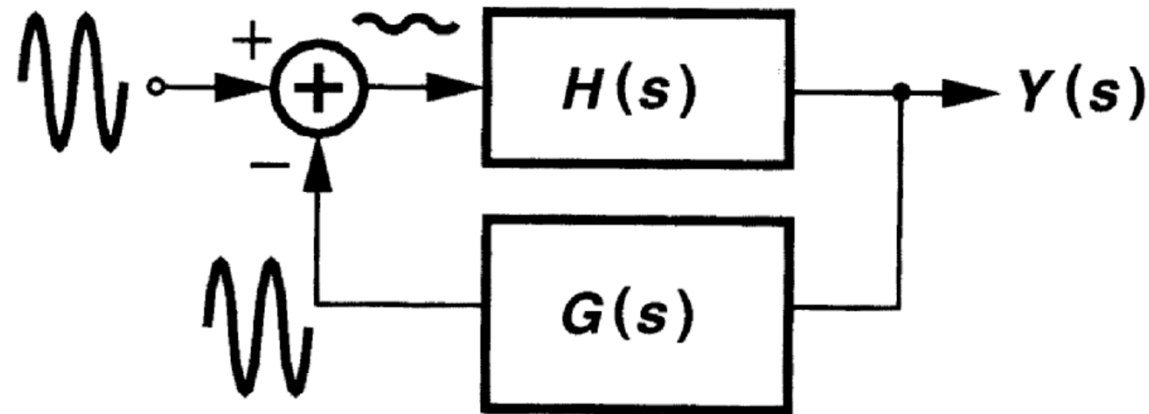
→

$$\frac{Y(s)}{X(s)} = \frac{H(s)}{1 + H(s)G(s)} = \frac{1}{G(s)} \frac{H(s)G(s)}{1 + H(s)G(s)}$$

- (1): A feedforward amplifier
- (2): A means to sense the output
- (3): The feedback network
- (4): A means to generate the feedback error



Signal at the input of $H(s)$



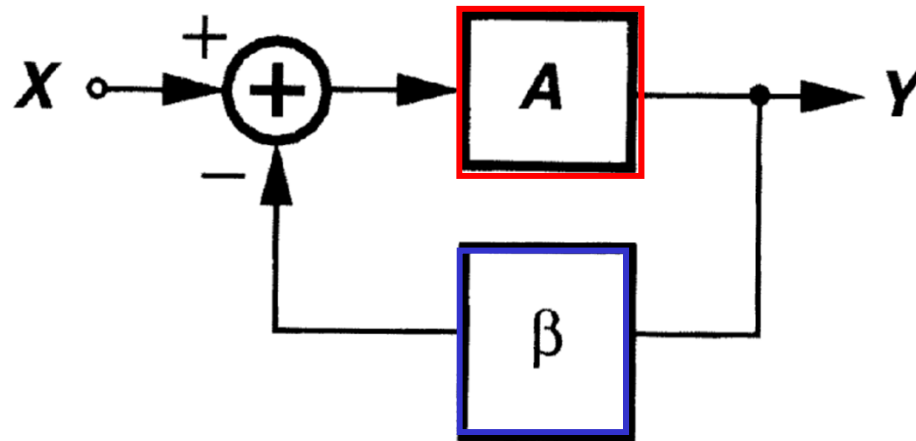
The signal amplitude at the input of $H(s)$ is very small

Often called the “virtual ground”

Outline

- **General consideration**
 - Abstract model
 - **Properties of feedback circuits**
- Feedback topologies
- Effect of feedback on noise
- Feedback analysis difficulties
- Two-port network and loading effect

(1) Gain Desensitization



A is large but sensitive to environment variations

$$A_{CL} = \frac{Y}{X} = \frac{A}{1 + A\beta} = \frac{1}{\beta} \cdot \frac{A\beta}{1 + A\beta} = \frac{1}{\beta} \left(1 - \frac{1}{1 + \beta A} \right) \approx \frac{1}{\beta} \left(1 - \frac{1}{\beta A} \right)$$

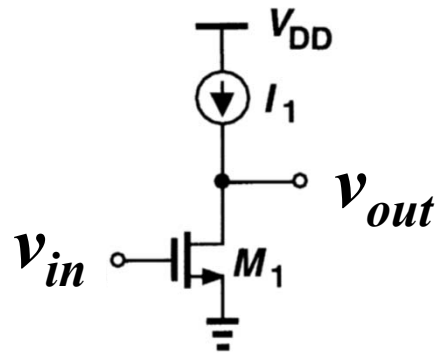
If βA (loop gain) is large enough

$\longrightarrow A_{CL} \approx \frac{1}{\beta}$

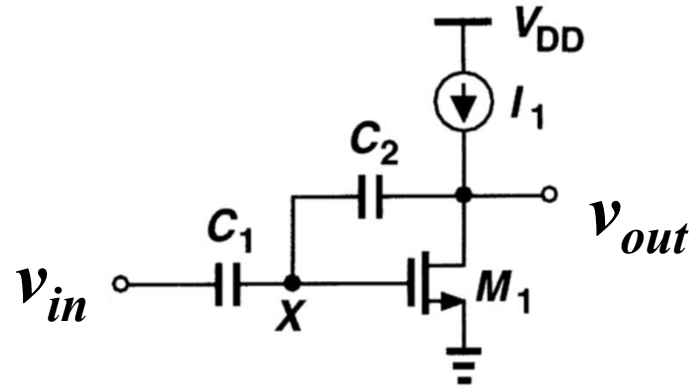
A_{CL} is much less sensitive to environment variations!!



Example of gain stabilization



$$\frac{v_{out}}{v_{in}} = -g_m r_{o1}$$



$$v_{out} = -g_m r_{o1} \cdot v_x$$

$$(v_{out} - v_x) \cdot sC_2 = (v_x - v_{in}) \cdot sC_1$$

$$\frac{v_{out}}{v_{in}} = - \frac{1}{\left(1 + \frac{1}{g_{m1} r_{O1}}\right) \frac{C_2}{C_1} + \frac{1}{g_{m1} r_{O1}}}$$

$$\approx -C_1 / C_2, \text{ for large } g_{m1} r_{O1}$$



Figuring out the loop gain

- Cut the loop at any point

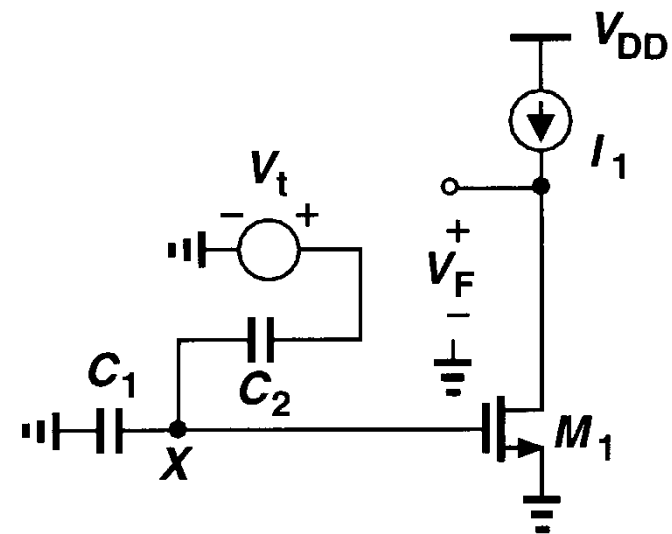
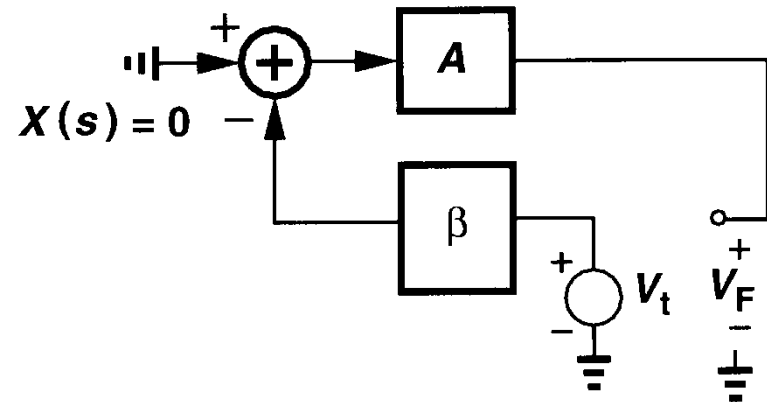
$$v_t \beta (-1) A = v_f$$

$$\rightarrow v_f / v_t = -\beta A$$

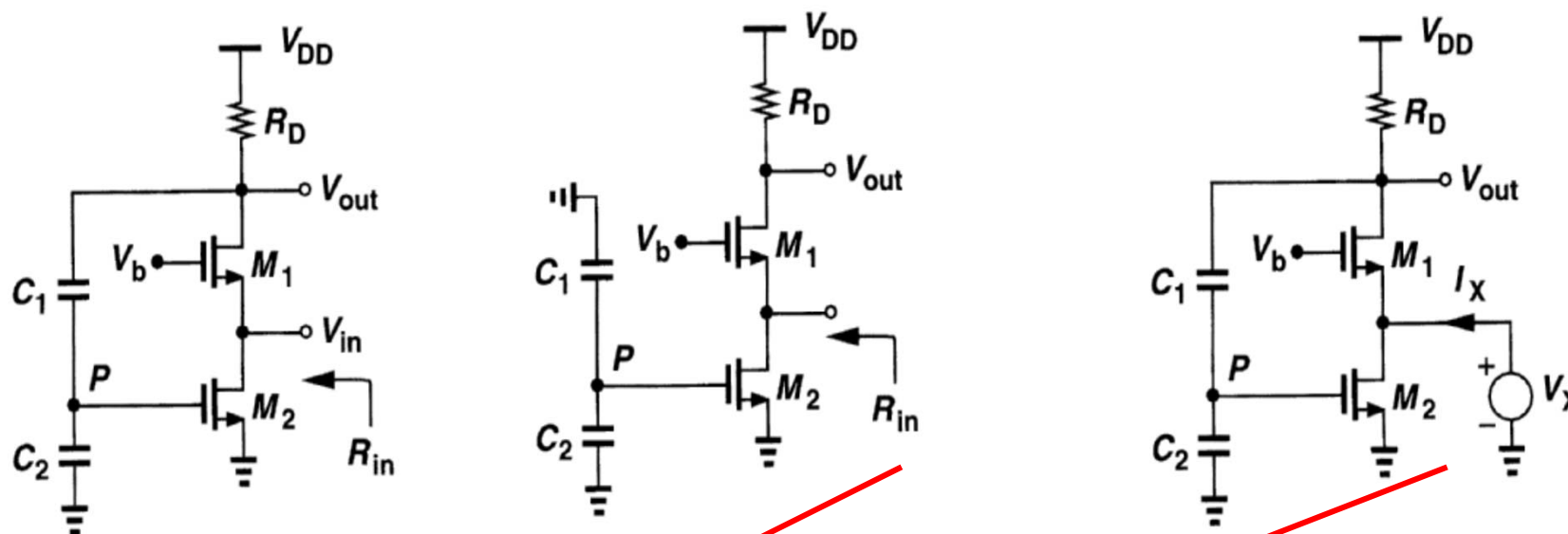
- For the above example

$$v_t \frac{C_2}{C_1 + C_2} (-g_m r_{o1}) = v_f$$

$$\rightarrow v_f / v_t = -\frac{C_2}{C_1 + C_2} g_m r_{o1}$$



(2) Terminal impedance modification: **input node**



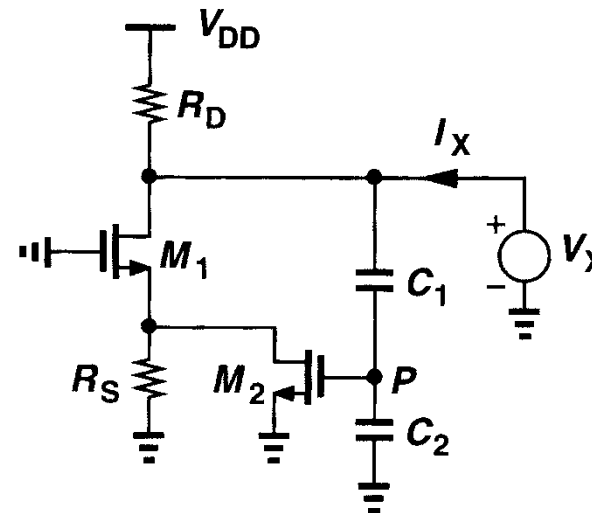
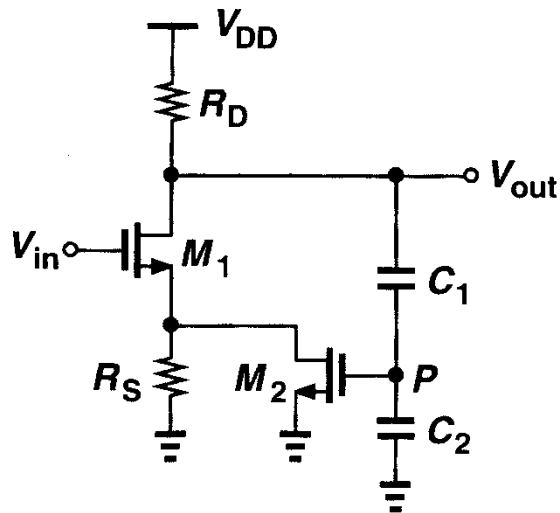
$$R_{in,open} = \frac{1}{g_{m1} + g_{mb1}}$$

$$R_{in,closed} = \frac{v_x}{i_x} = \frac{1}{g_{m1} + g_{mb1}} \frac{1}{1 + \boxed{g_{m2} R_D \frac{C_1}{C_1 + C_2}}}$$

Loop gain

**Input impedance
is reduced by
feedback**

(2) Terminal Impedance Modification: **output node**



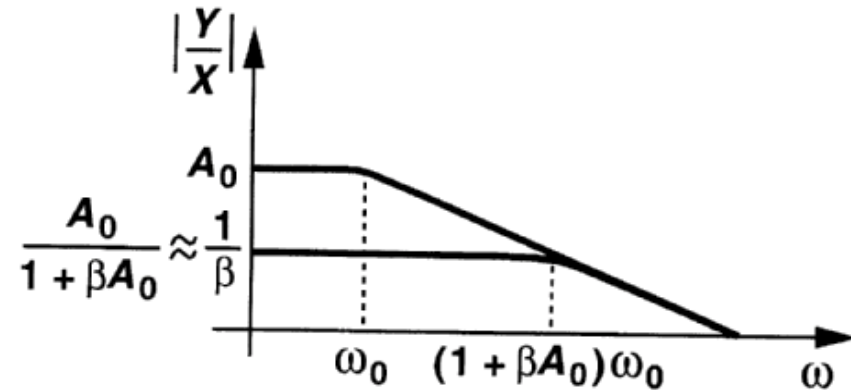
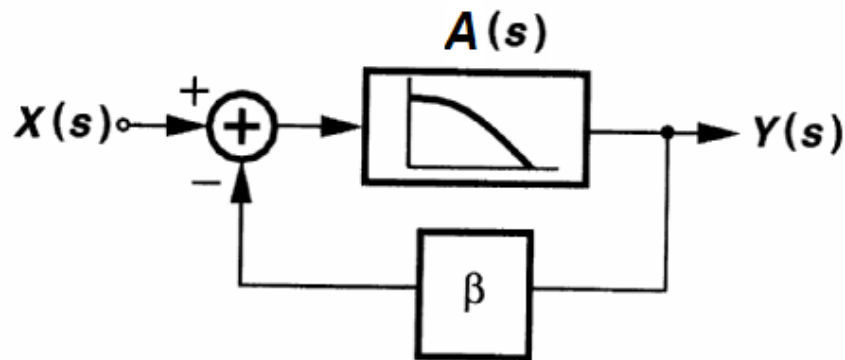
$$R_{out,open} = R_D$$

Output impedance is reduced by feedback

$$R_{out,closed} = \frac{v_x}{i_x} = \frac{R_D}{1 + \underbrace{\frac{C_1}{C_1 + C_2} \cdot \frac{g_{m2} R_S}{1 + (g_{m1} + g_{mb1}) R_S} \cdot (g_{m1} + g_{mb1}) R_D}_{\text{Loop gain}}}$$



(3) Bandwidth Modification

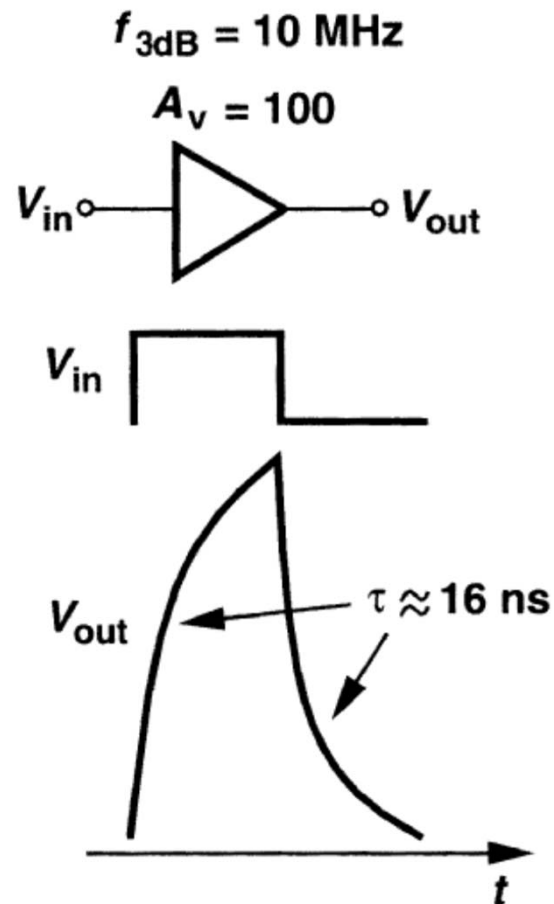


$$A(s) = \frac{A_0}{1 + \frac{s}{\omega_0}} \quad \frac{Y}{X}(s) = \frac{\frac{A_0}{1 + \frac{s}{\omega_0}}}{1 + \beta \frac{A_0}{1 + \frac{s}{\omega_0}}} = \frac{A_0}{1 + \beta A_0 + \frac{s}{\omega_0}} = \frac{\frac{A_0}{1 + \beta A_0}}{1 + \frac{s}{(1 + \beta A_0)\omega_0}}$$

The gain is decreased by $(1 + \beta A_0)$

The pole frequency is increased by $(1 + \beta A_0)$

How feedback improves speed?



(a)

To amplify a 20MHz impulse
with a gain of 100

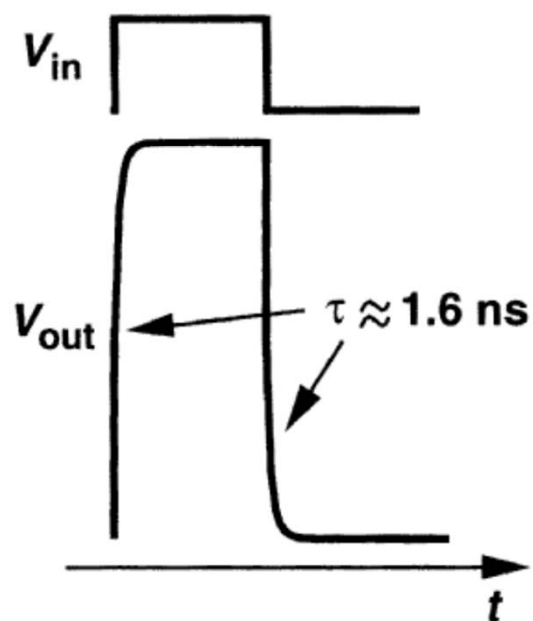
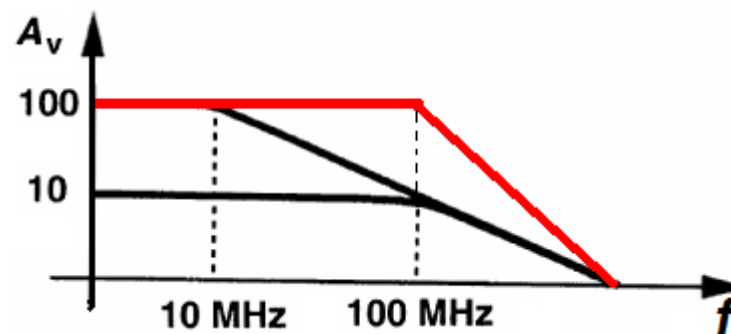
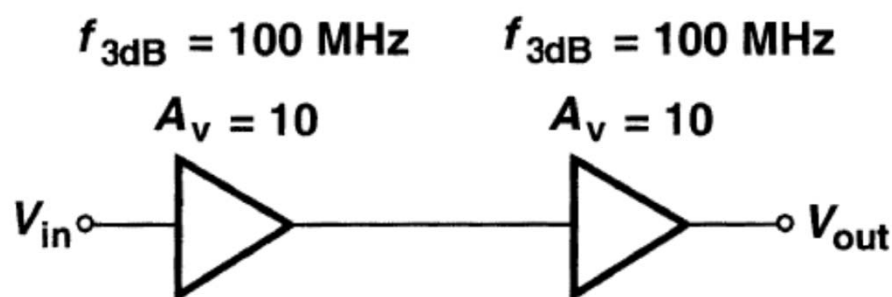
Method 1:

Use a amplifier with gain of 100
But the band width is only 10MHz

Rise time and faltime are too long



How feedback improves speed?



Method 2:

Use feedback to obtain a bandwidth of 100MHz

The gain is also reduced to 10

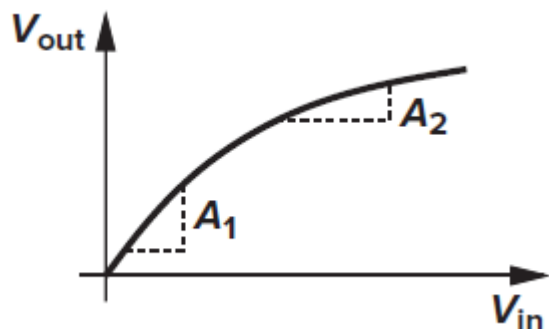
Cascade 2 stages

The speed is much improved!!



Suppression of Nonlinearity

- Before feedback

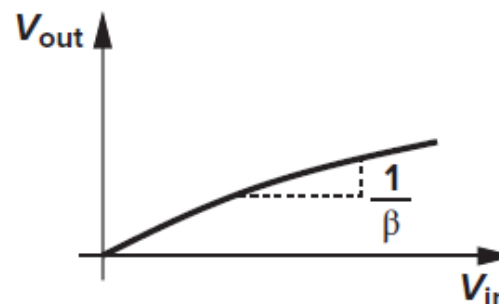


Gain ratio between different inputs

$$r_{open} = \frac{A_2}{A_1}$$

$$r_{open} = 1 - \frac{\Delta A}{A_1}$$

- After feedback



Gain ratio between different inputs

$$r_{closed} = \frac{\frac{A_2}{1 + \beta A_2}}{\frac{A_1}{1 + \beta A_1}}$$

$$r_{closed} \approx 1 - \frac{\frac{1}{\beta A_2} - \frac{1}{\beta A_1}}{1 + \frac{1}{\beta A_2}}$$

$$\approx 1 - \frac{A_1 - A_2}{1 + \beta A_2} \frac{1}{A_1}$$

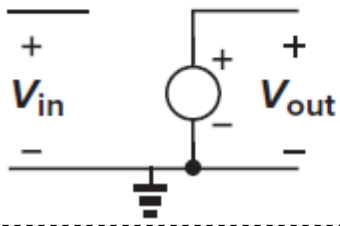
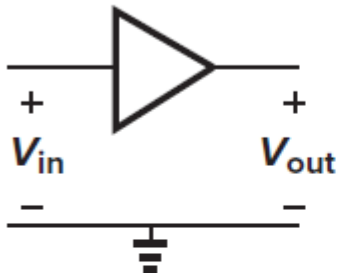
$$= \frac{1 + \frac{1}{\beta A_1}}{1 + \frac{1}{\beta A_2}}$$

$$\approx 1 - \frac{\Delta A}{1 + \beta A_2} \frac{1}{A_1}$$

Lecture #19

Types of Amplifiers

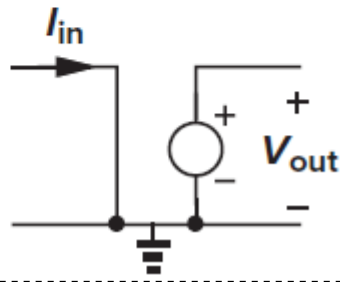
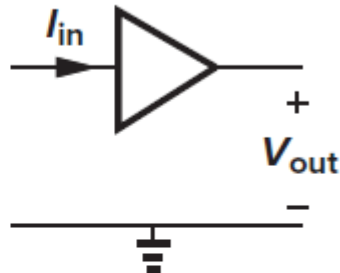
Voltage Amp.



$$Z_{in} = \infty$$

$$Z_{out} = 0$$

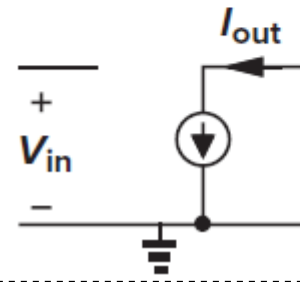
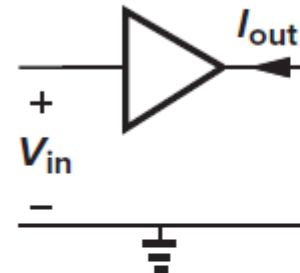
Transimpedance Amp.



$$Z_{in} = 0$$

$$Z_{out} = 0$$

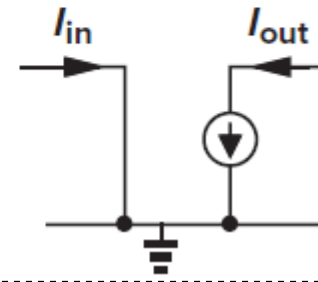
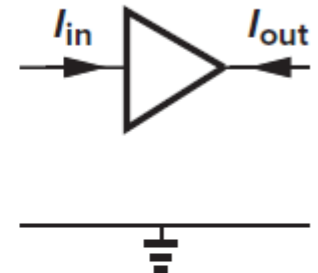
Transconductance Amp.



$$Z_{in} = \infty$$

$$Z_{out} = \infty$$

Current Amp.



$$Z_{in} = 0$$

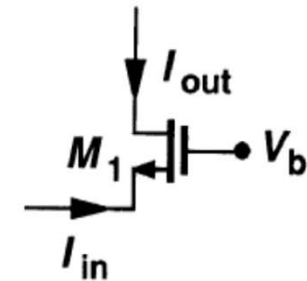
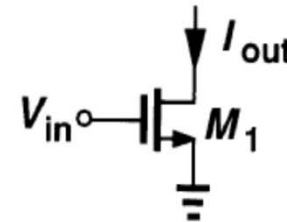
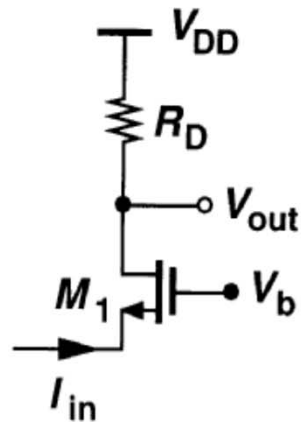
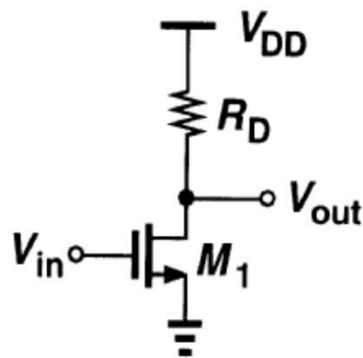
$$Z_{out} = \infty$$

Typical CMOS amplifier

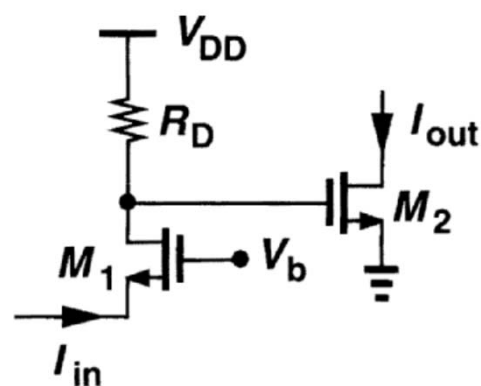
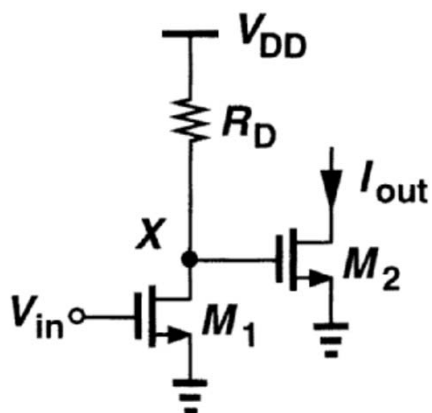
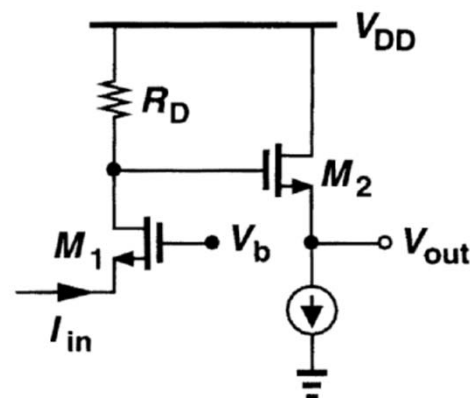
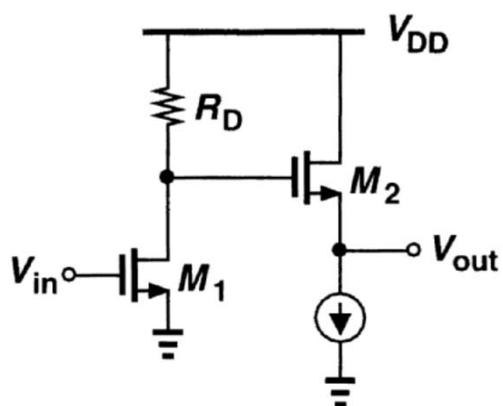


Types of Amplifiers (Cont.)

Simple implementation of four types of amplifiers

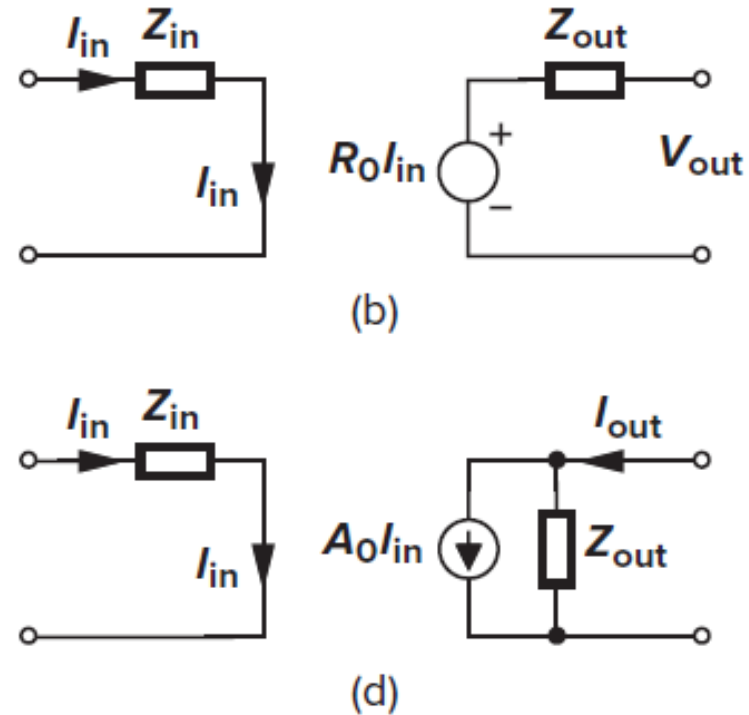
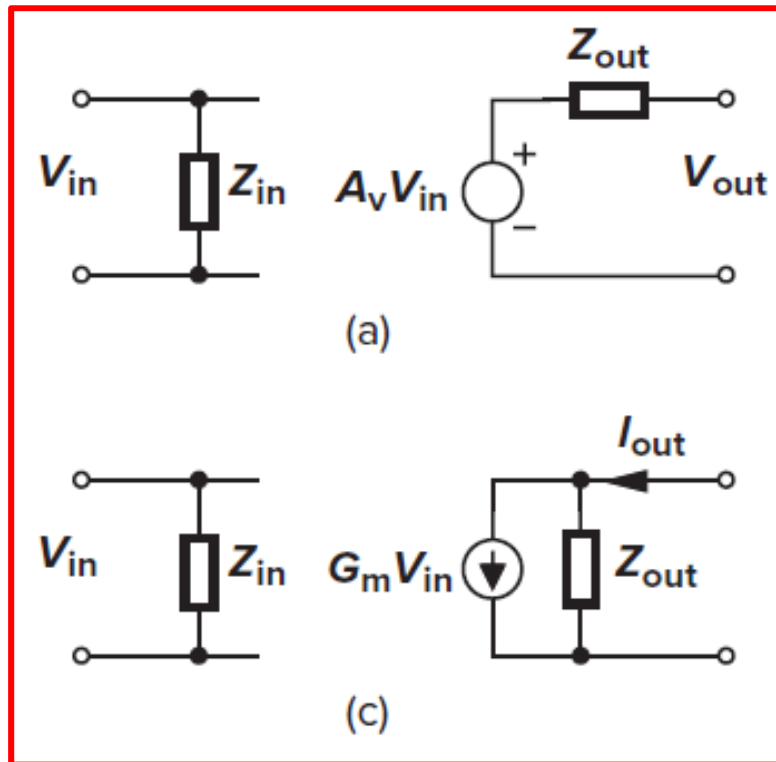


Examples of Improved Amplifiers



Non-ideal amplifier models

- With finite gain and input/output impedance



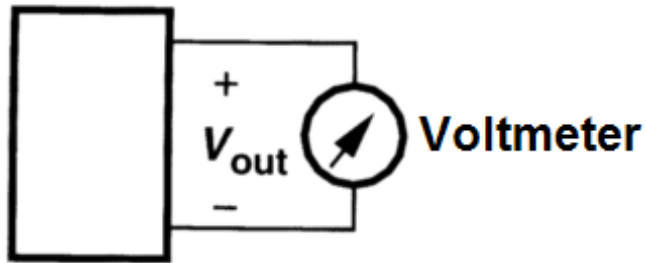
Sense and Feedback Mechanisms

- A sensing circuit is needed in feedback loops
 - Sense output signal and return it to the input
 - There are 4 types:
 - (1) Voltage — Voltage;
 - (2) Current — Voltage;
 - (3) Current — Current;
 - (4) Voltage — Current.

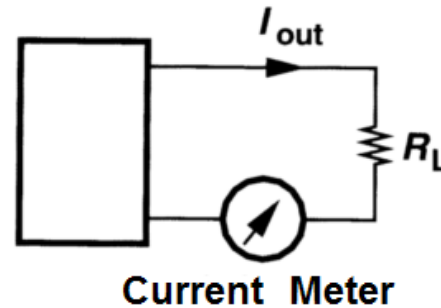


Sensing and adding methods

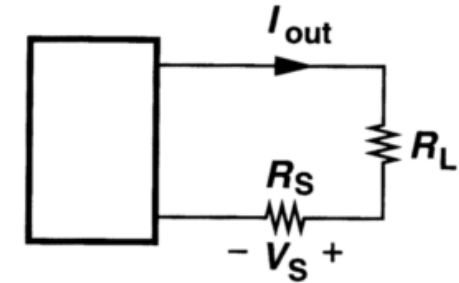
- Sensing voltage/ current



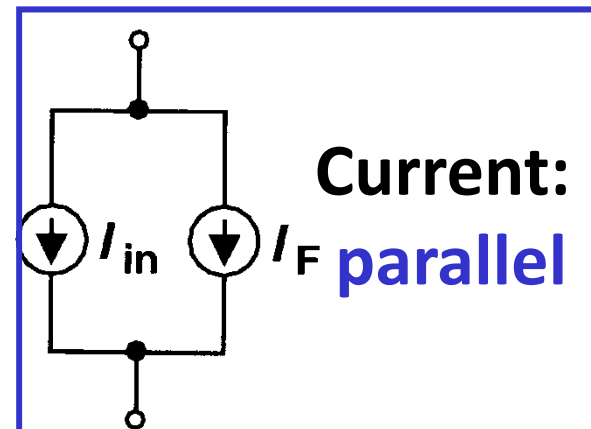
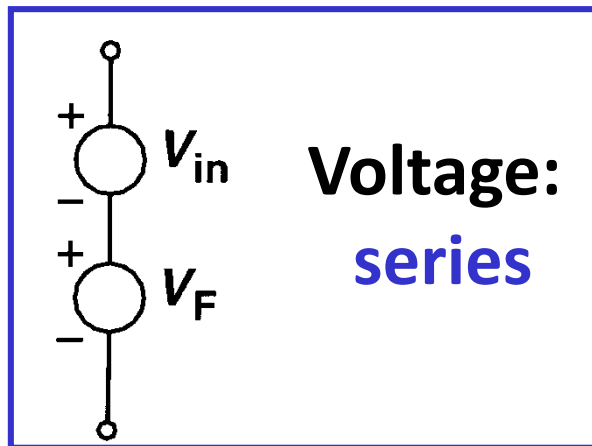
Voltage: **parallel**



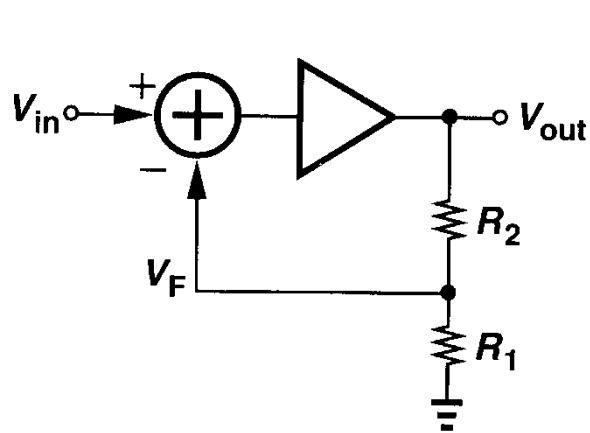
Current: **series**



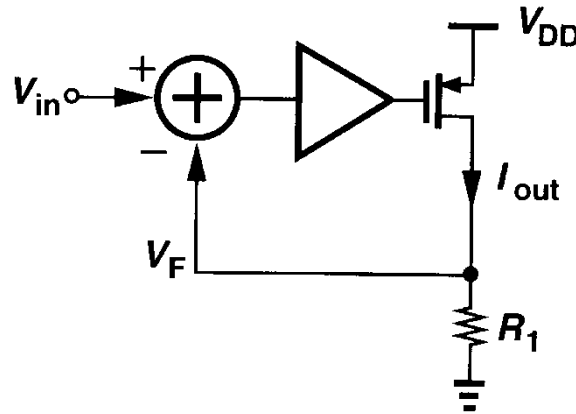
- Adding method



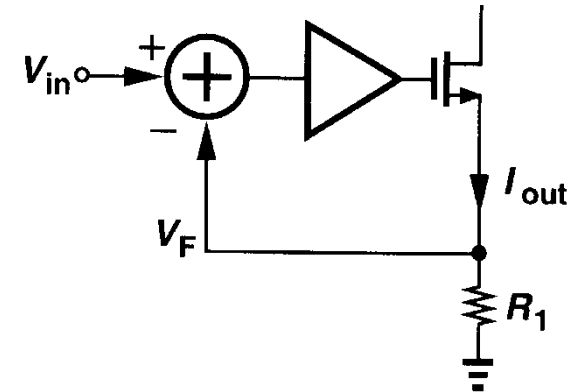
Practical implementations



- Sensing voltage and returning a voltage

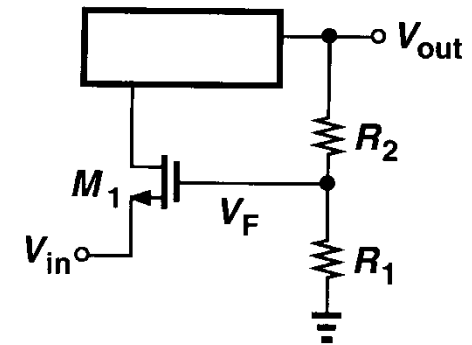
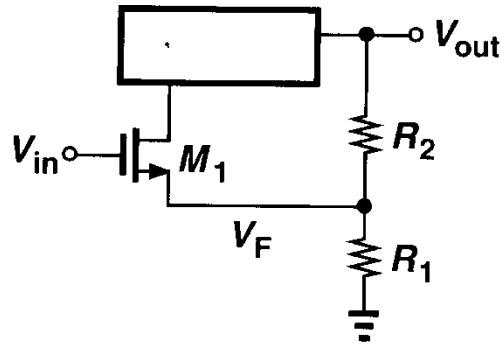
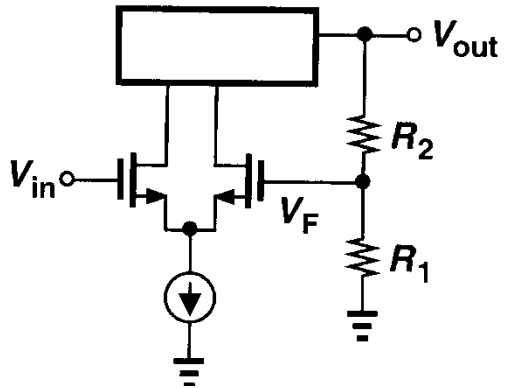


- Sensing current and returning a voltage

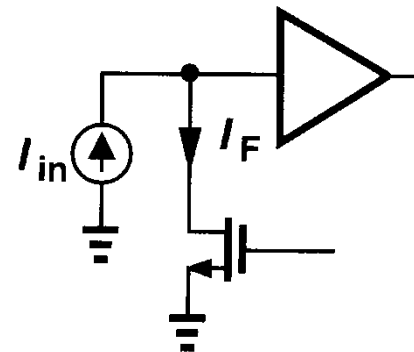
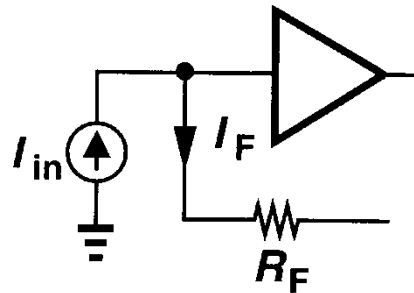


Adding methods

- Voltage adding



- Current adding



Outline

- General consideration
- **Feedback topologies**
 - **Voltage-Voltage Feedback**
 - Current-Voltage Feedback
 - Voltage-Current Feedback
 - Current-Current Feedback
- Effect of feedback on noise
- Feedback analysis difficulties
- Two-port network and loading effect

Voltage-Voltage Feedback

- Feedback network requirements:

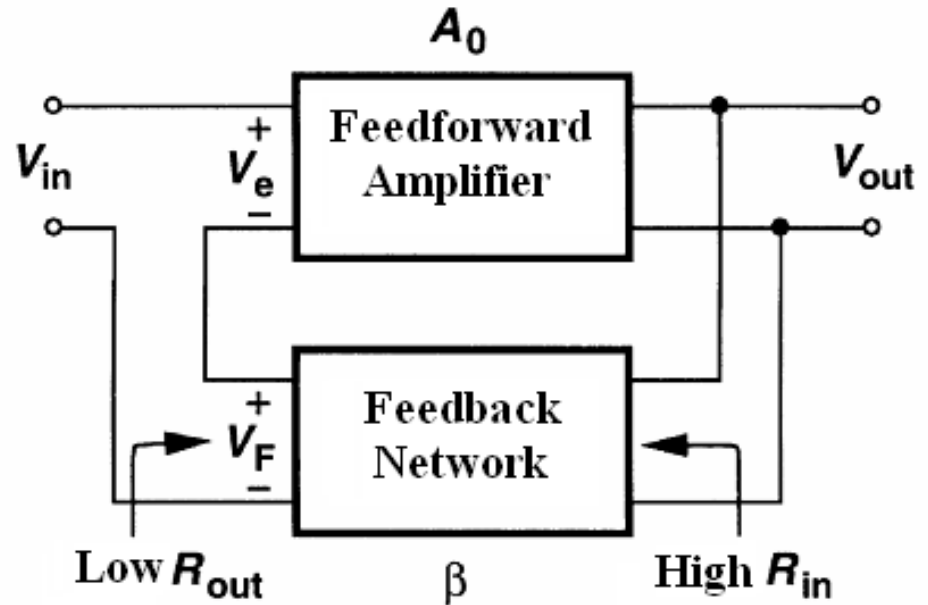
- Parallel with output
- Series with input
- **High R_{in} , Low R_{out}**

$$V_F = \beta V_{out}$$

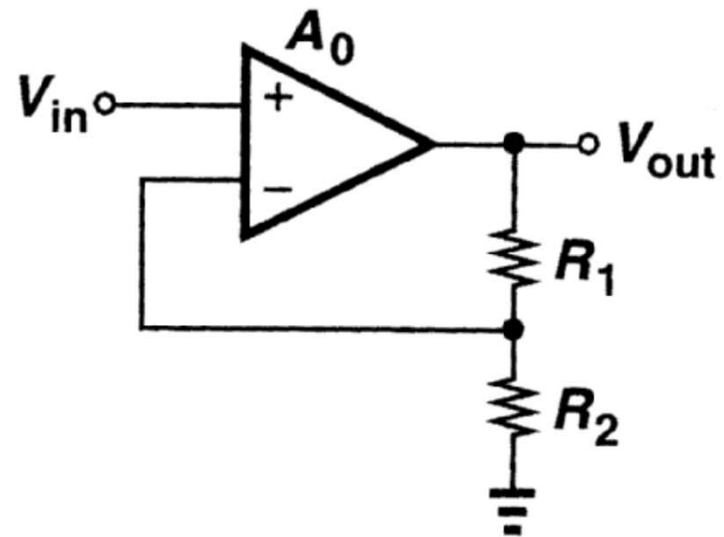
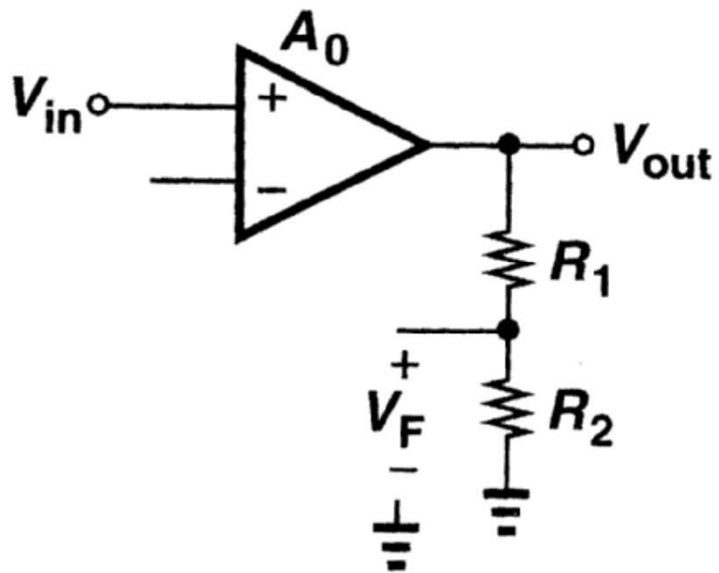
$$V_e = V_{in} - V_F = V_{in} - \beta V_{out}$$

$$V_{out} = A_0 V_e = A_0 (V_{in} - \beta V_{out})$$

$$\rightarrow A_{0,CL} = \frac{V_{out}}{V_{in}} = \frac{A_0}{1 + \beta A_0} = \frac{1}{\beta} \frac{\beta A_0}{1 + \beta A_0}$$

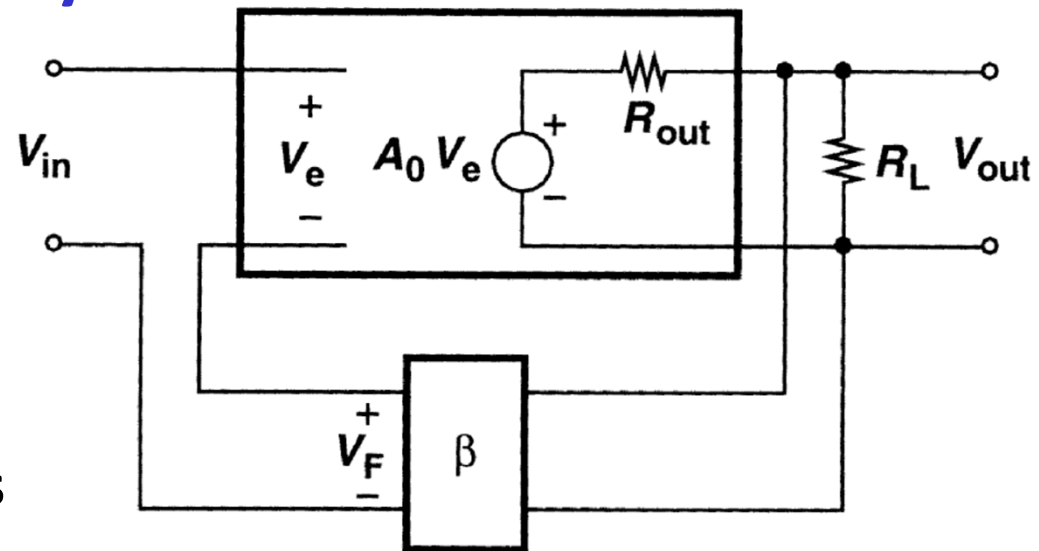


An example of Voltage-Voltage Feedback



Output Resistance: Observation

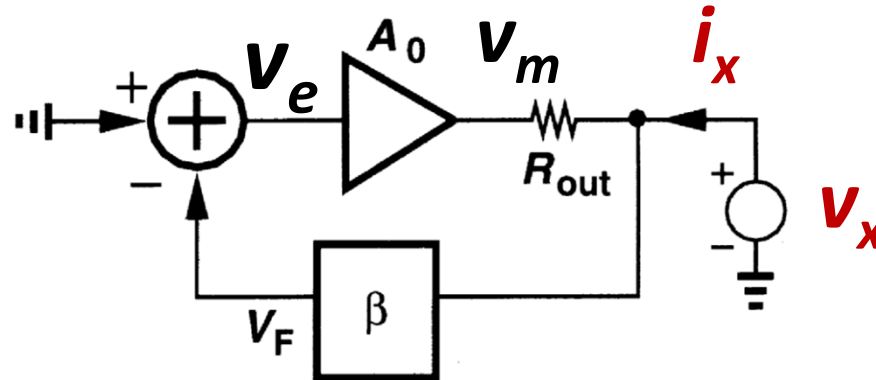
- If the network drives the load R_L ,
 R_L decreases gradually



- Open loop gain drops
- As long as loop gain $\gg 1$
 - Closed-loop gain keeps almost the same
 - The circuit acts like a voltage source
 - Exhibiting low output resistance



Output Resistance: calculation



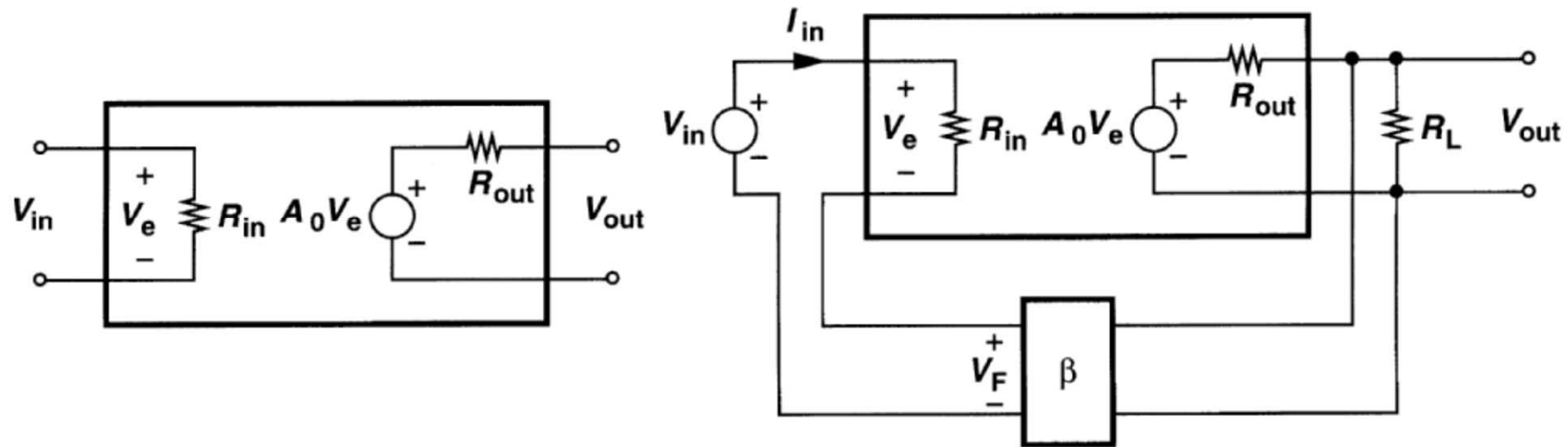
$$i_x = \frac{v_x - v_m}{R_{out}} = \frac{v_x - (-\beta v_x A_0)}{R_{out}} = \frac{v_x (1 + \beta A_0)}{R_{out}}$$

$$R_{out,CL} = \frac{v_x}{i_x} = \frac{R_{out}}{1 + \beta A_0}$$

R_{out} is reduced!!



Input Resistance: observation

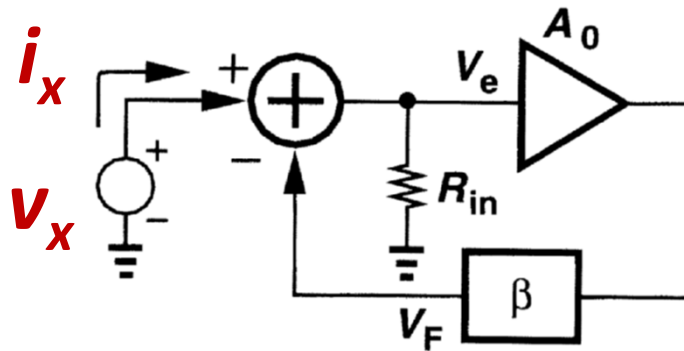


In open loop configuration, R_{in} sustains V_{in}

In closed loop: R_{in} sustain only a fraction of V_{in}

→ I_{in} is reduced, meaning that R_{in} is increased

Input Resistance : calculation



$$v_e = i_X R_{in}$$

$$v_e = v_X - v_F = v_X - \beta A_0 i_X R_{in}$$

$$i_X R_{in} = v_X - \beta A_0 i_X R_{in}$$

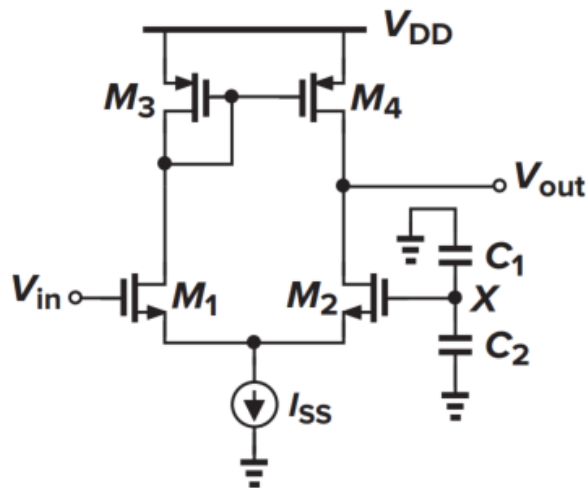
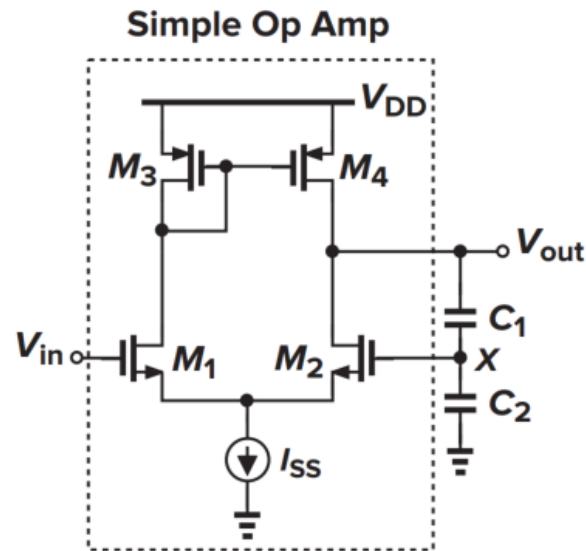
$$i_X R_{in} (1 + \beta A_0) = v_X$$

$$R_{in,CL} = \frac{v_X}{i_X} = R_{in} (1 + \beta A_0)$$



Ex. 8.4

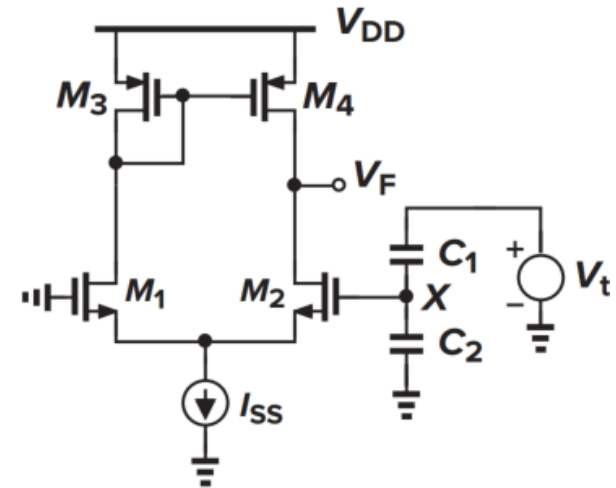
- A 5-T OTA configured in non-inverting amplifier with capacitive feedback. Calculate **low-frequency** $A_{V,CL}$ and $R_{out,CL}$



Open-loop gain = ?

$$g_{m1}(r_{O2} \parallel r_{O4})$$

A_0



Loop gain = ?

$$\beta A_0 = \frac{C_1}{C_1 + C_2} \boxed{g_{m1}(r_{O2} \parallel r_{O4})}$$

β A_0



Ex. 8.4 (Cont')

$$A_{V,CL} = \frac{A_0}{1 + \beta A_0} = \frac{1}{\beta} \frac{\beta A_0}{1 + \beta A_0} = \frac{C_1 + C_2}{C_1} \frac{\frac{C_1}{C_1 + C_2} g_{m1} (r_{o2} \parallel r_{o4})}{1 + \frac{C_1}{C_1 + C_2} g_{m1} (r_{o2} \parallel r_{o4})}$$

$$\text{If } \beta A_0 \gg 1, \quad A_{V,CL} \approx \frac{1}{\beta} = 1 + \frac{C_2}{C_1}$$

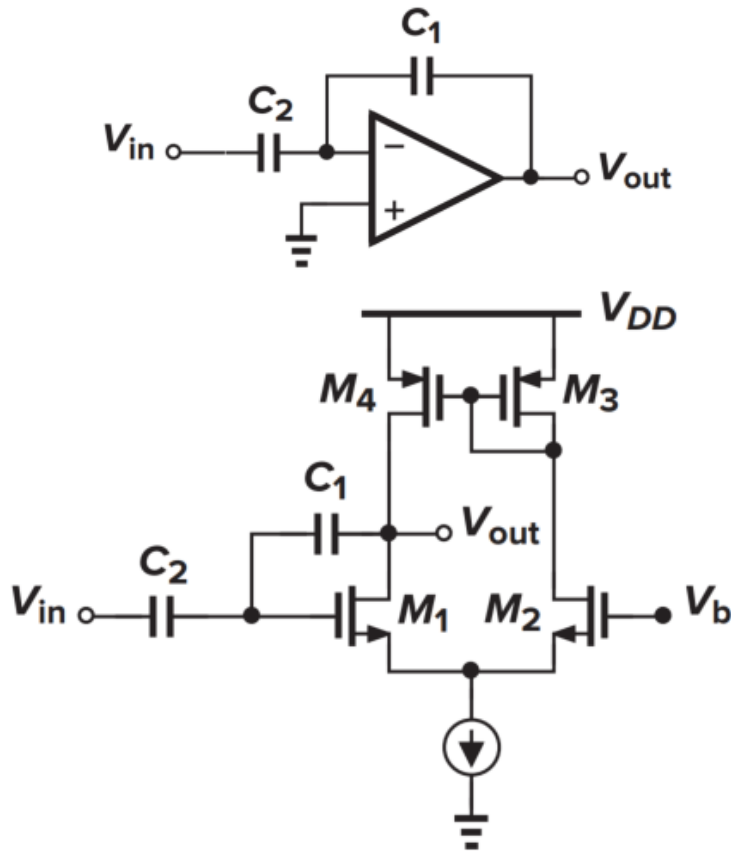
$$R_{out,open} = r_{o2} \parallel r_{o4} \rightarrow R_{out,CL} = R_{out,open} / (1 + \beta A_0)$$

$$R_{out,CL} = \frac{r_{o2} \parallel r_{o4}}{1 + \frac{C_1}{C_1 + C_2} g_{m1} (r_{o2} \parallel r_{o4})} \approx \left(1 + \frac{C_2}{C_1} \right) \frac{1}{g_{m1}}$$



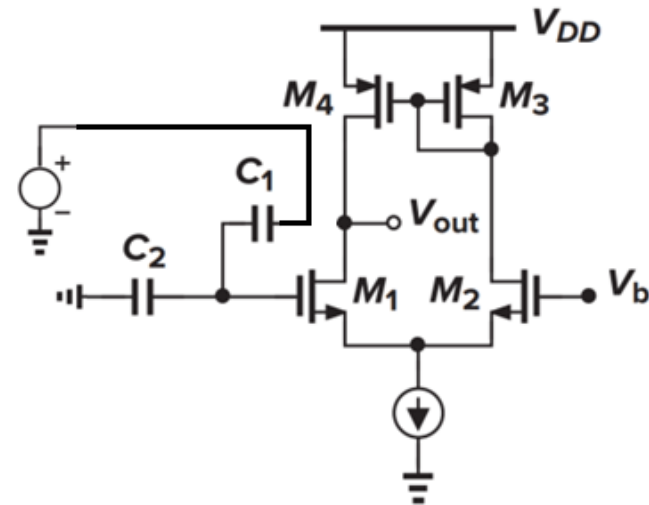
Ex. 8.5

- A 5-T OTA configured in **inverting** amplifier with capacitive feedback. Calculate **low-frequency** $A_{V,CL}$ and $R_{out,CL}$



Loop Gain = ?

With V_{in} set to zero

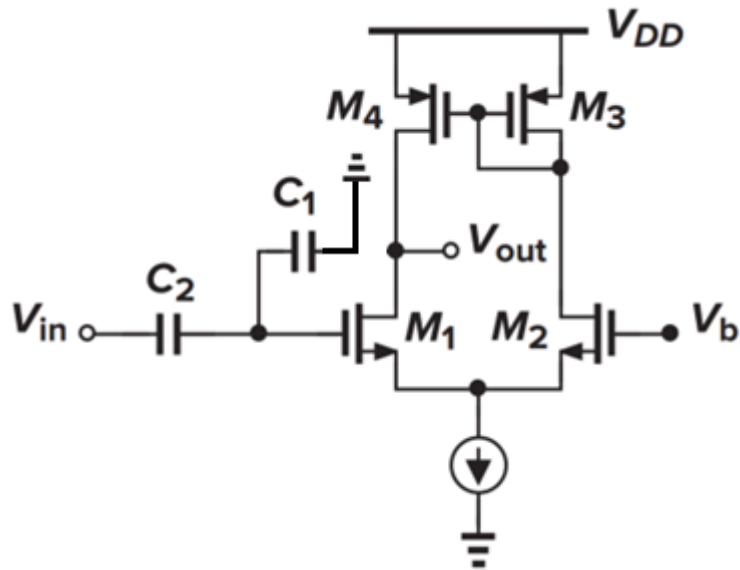


Loop gain is the same as Ex. 8.4

$$\beta A_0 = \frac{C_1}{C_1 + C_2} g_{m1} (r_{O1} \parallel r_{O4})$$



Ex. 8.5 (Cont')



$$R_{\text{out,open}} = r_{o2} \parallel r_{o4}$$

$$R_{\text{out,CL}} = R_{\text{out,open}} / (1 + \beta A_0)$$

$R_{\text{out,CL}}$ is the same as Ex. 8.4

Open loop gain

$$A_{\text{open}} = \frac{C_2}{C_1 + C_2} A_0 = \frac{C_2}{C_1 + C_2} g_{m1} (r_{o1} \parallel r_{o4})$$

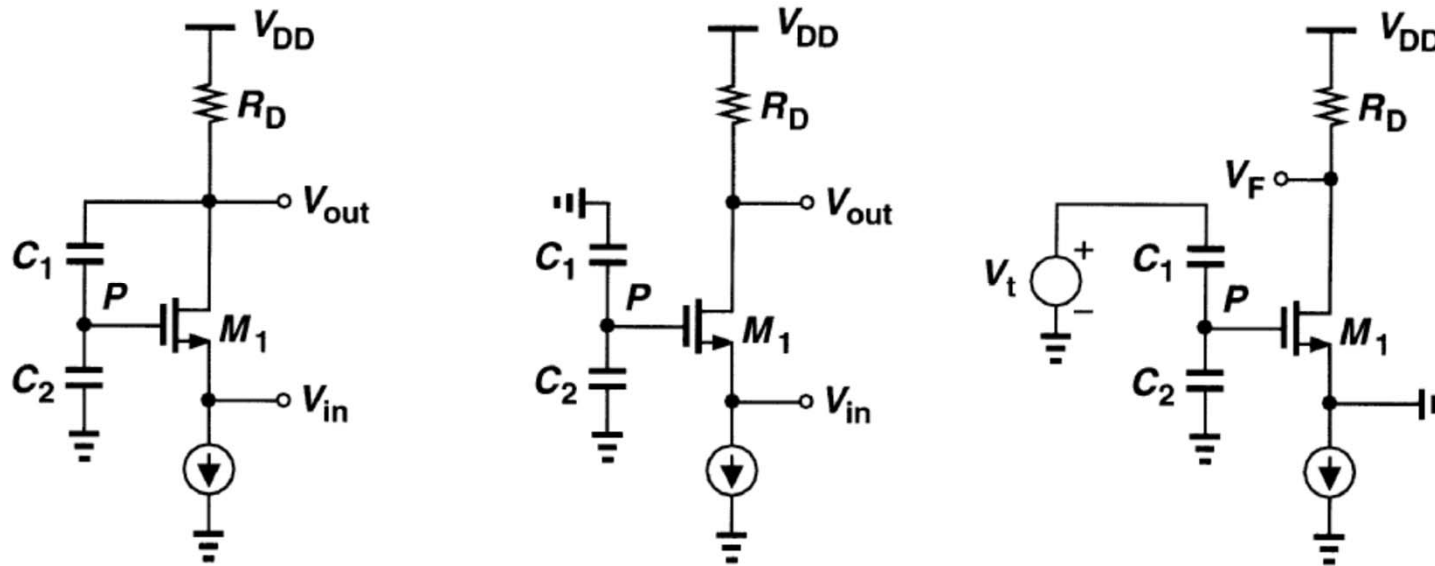
Closed-loop gain

$$A_{V,CL} = \frac{\frac{C_2}{C_1 + C_2} g_{m1} (r_{o2} \parallel r_{o4})}{1 + \frac{C_1}{C_1 + C_2} g_{m1} (r_{o2} \parallel r_{o4})} = \frac{C_2}{C_1} \frac{\beta A_0}{1 + \beta A_0}$$

Note the difference with Ex. 8.4



Ex. 8.5: Input resistance in VVF



$$R_{in,CL} = \frac{V_X}{I_X} = R_{in} (1 + \beta A_0)$$

$$\beta A_0 = \frac{C_1}{C_1 + C_2} g_{m1} R_D$$

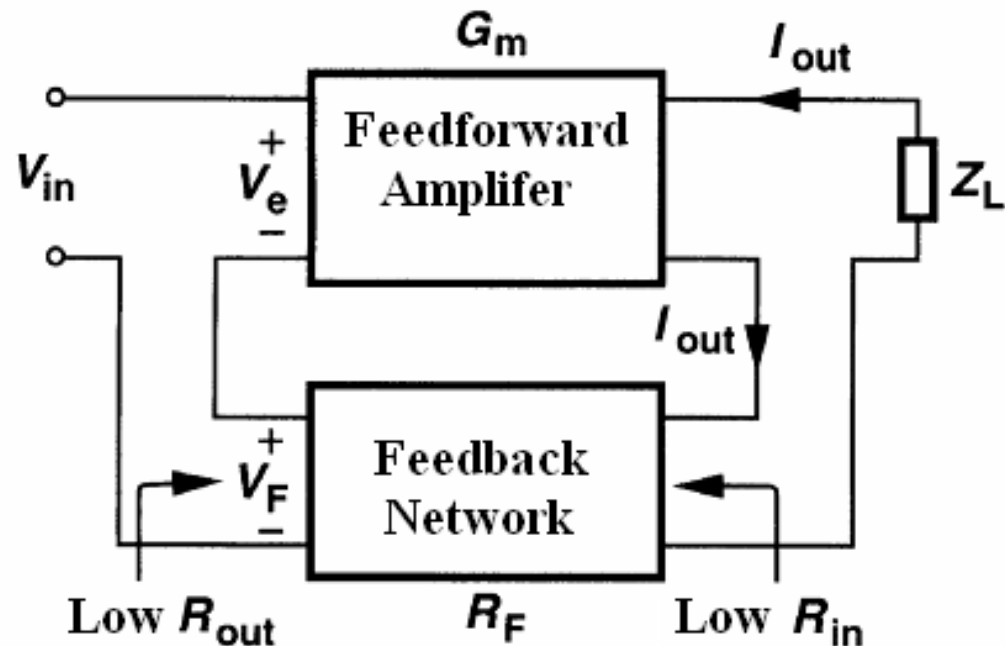
$$\longrightarrow R_{in,closed} = \frac{1}{g_{m1} + g_{mb1}} \left(1 + \frac{C_1}{C_1 + C_2} g_{m1} R_D \right)$$

Outline

- General consideration
- Feedback topologies
 - Voltage-Voltage Feedback
 - **Current-Voltage Feedback**
 - Voltage-Current Feedback
 - Current-Current Feedback
- Effect of feedback on noise
- Feedback analysis difficulties

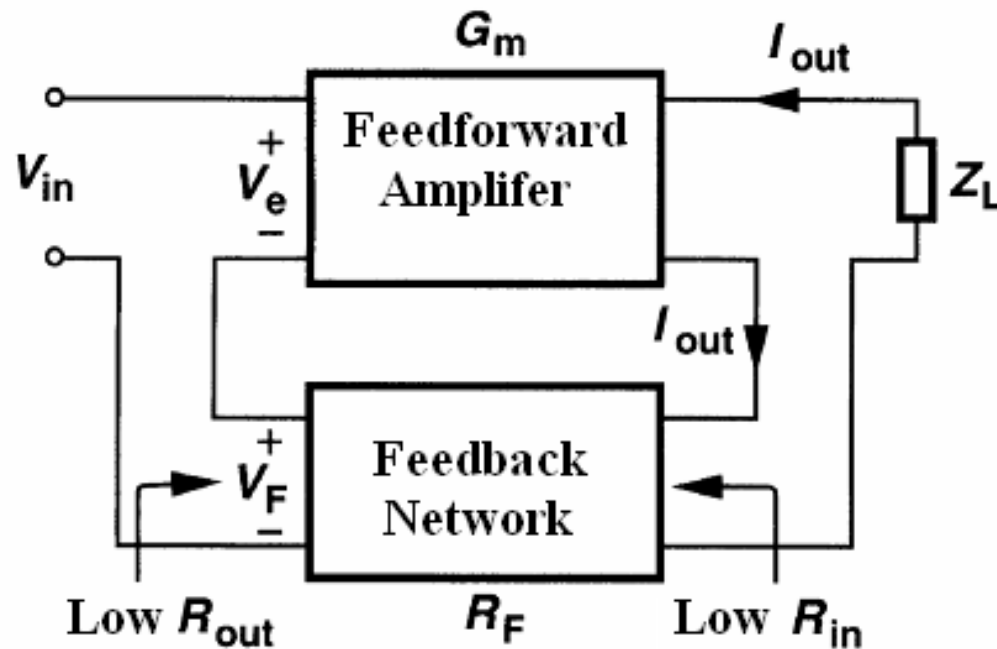
Current-Voltage Feedback

- The feedforward amplifier is an OTA with gain of G_m
- Sense current and return voltage
- Feedback network requirements:
 - Series with output
 - Series with input
 - Low R_{in}
 - Low R_{out}



Calculating the closed-loop transfer function

- The feedback circuit has a current input and a voltage output
- Denoting it with resistor R_F



$$I_{out} = G_m V_e = G_m (V_{in} - V_F)$$

$$= G_m (V_{in} - R_F I_{out})$$

$$I_{out} (1 + G_m R_F) = G_m V_{in}$$

$$G_{m,CL} = \frac{I_{out}}{V_{in}} = \frac{G_m}{1 + G_m R_F}$$

$$= \frac{1}{R_F} \frac{G_m R_F}{1 + G_m R_F}$$

Verifying the loop gain

$$G_{m,CL} = \frac{I_{out}}{V_{in}} = \frac{1}{R_F} \frac{G_m R_F}{1 + \boxed{G_m R_F}}$$

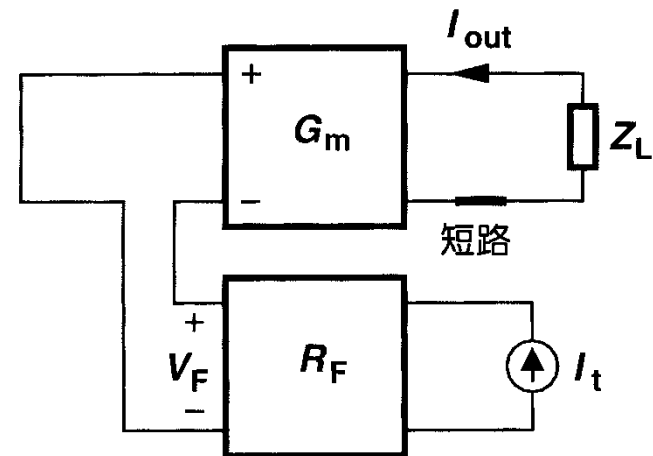
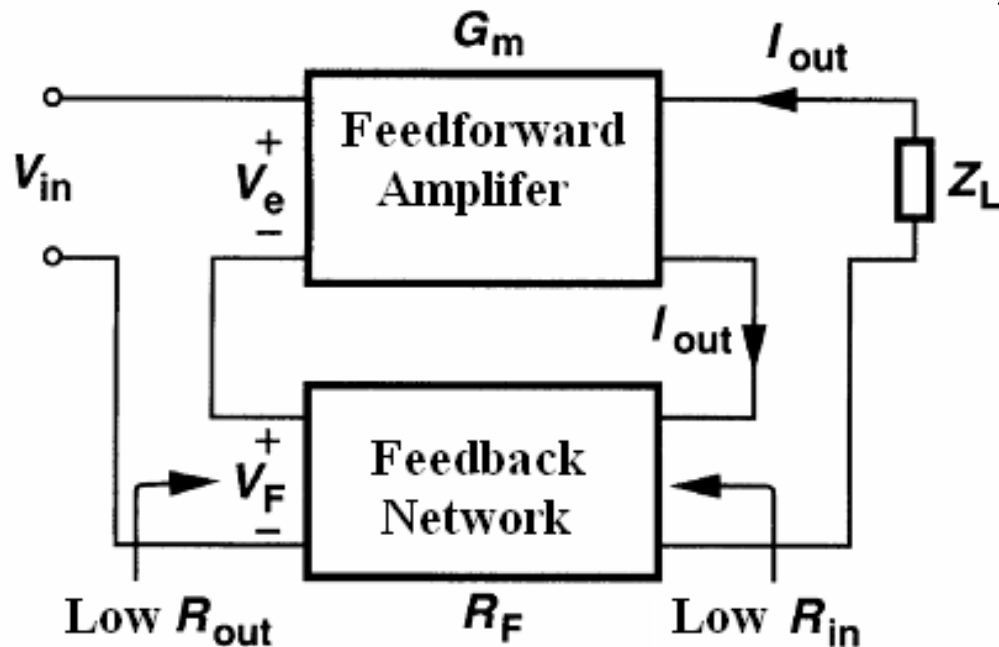
Loop gain

- Cut the loop

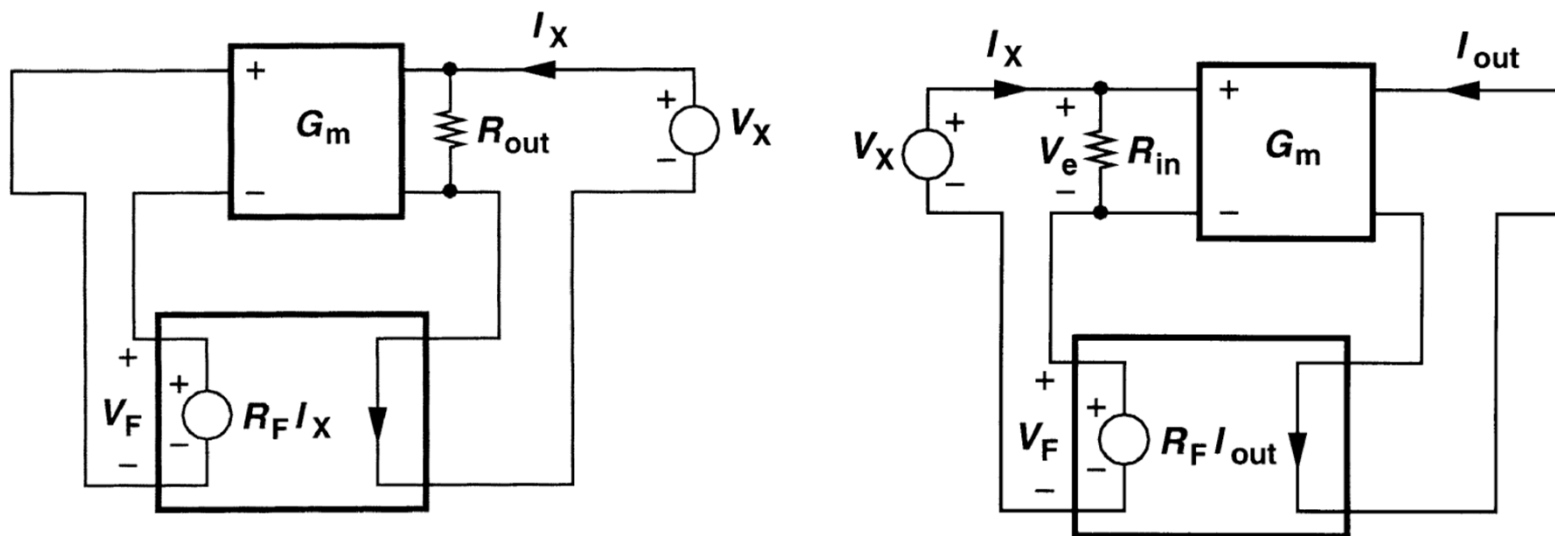
- Shorting the input and output

$$V_F = I_t R_F$$

$$I_{out} = -G_m V_F = -I_t \cdot R_F G_m$$



Output and input resistance in CVF

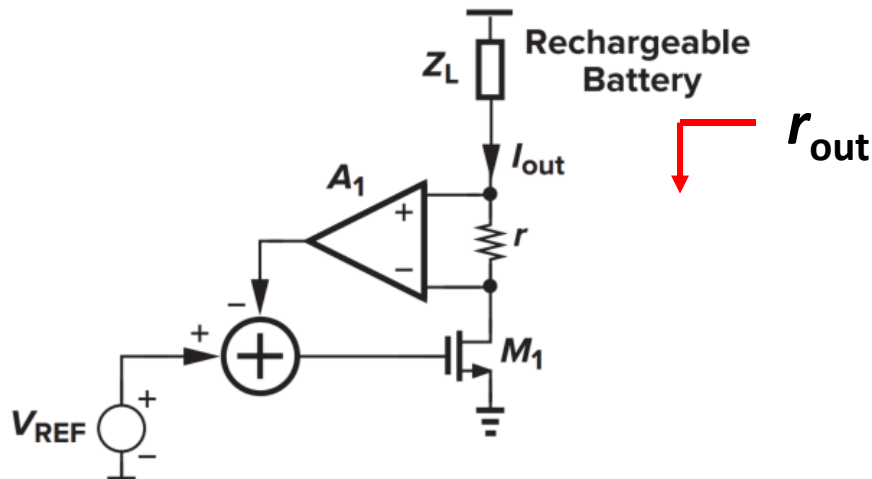


$$R_{out,CL} = \frac{V_X}{I_X} = R_{out} (1 + G_m R_F) \quad R_{in,CL} = \frac{V_X}{I_X} = R_{in} (1 + G_m R_F)$$

- Both R_{in} and R_{out} are increased!!

Ex. 8.7

- Charging circuit with **constant current**, calculate I_{out} and r_{out}
 - Assuming a high loop gain



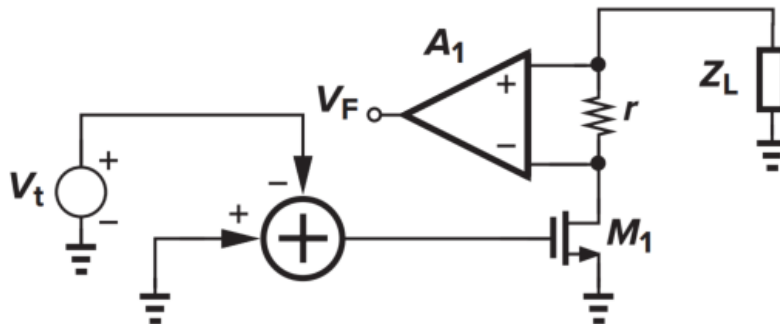
$$I_{out} r \cdot A_1 \approx V_{REF}$$

$$\rightarrow I_{out} \approx \frac{V_{REF}}{r \cdot A_1}$$

- Loop gain=?

$$\frac{V_F}{V_t} \approx -g_m r A_1$$

$$R_{out, closed} = (1 + g_m r A_1)(r_O + r)$$



A better charge current is obtained!



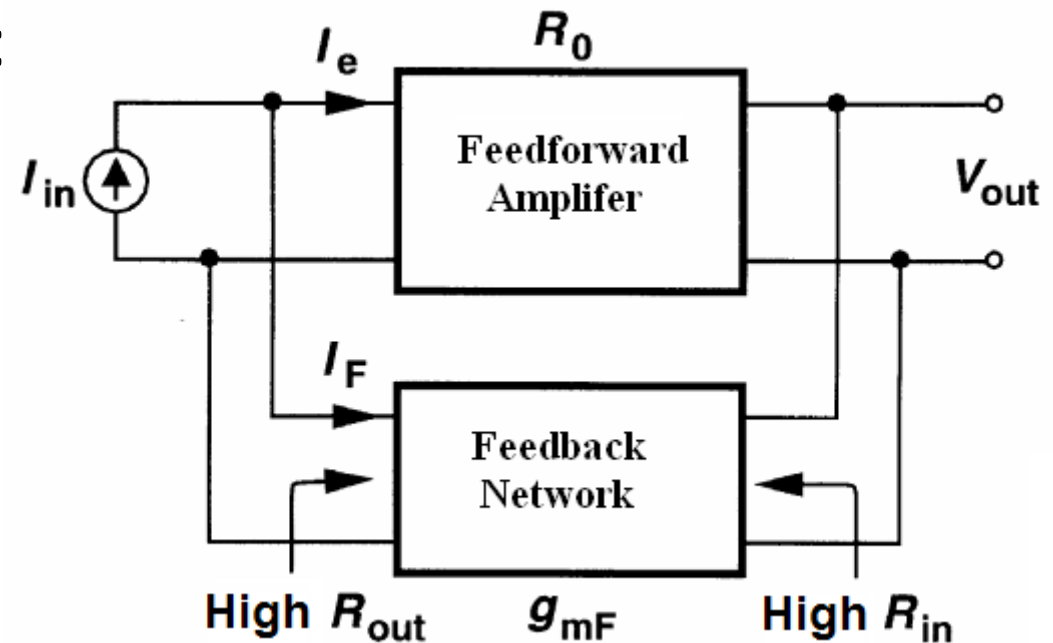
Outline

- General consideration
- Feedback topologies
 - Voltage-Voltage Feedback
 - Current-Voltage Feedback
 - **Voltage-Current Feedback**
 - Current-Current Feedback

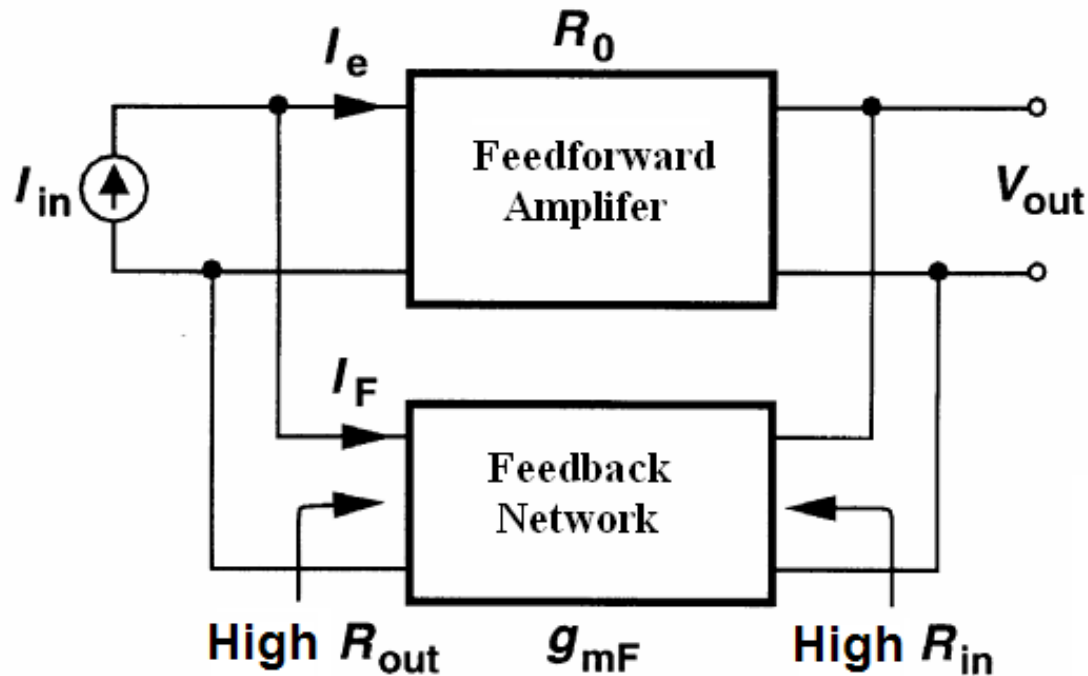


8.2.3 Voltage - Current Feedback

- The feedforward amplifier is an current-mode amplifier with a gain of R_0
- Sense voltage and return current
- Feedback network requirements:
 - Parallel with output
 - Parallel with input
 - High R_{in}
 - Low R_{out}



Closed-loop transfer function



$$I_F = g_{mF} V_{out}$$

$$I_e = I_{in} - g_{mF} V_{out}$$

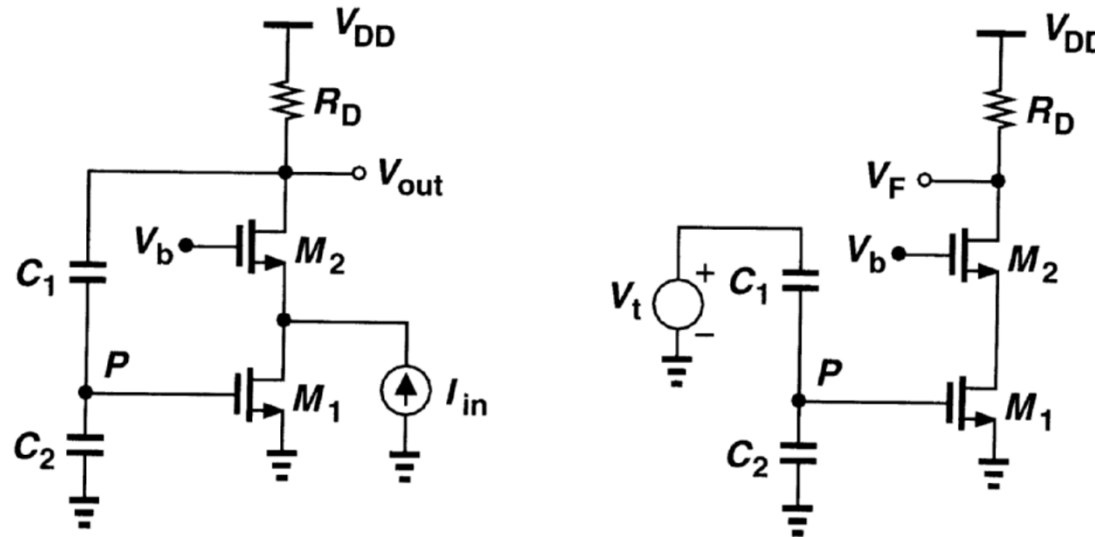
$$V_{out} = I_e R_0 = (I_{in} - g_{mF} V_{out}) R_0$$

$$R_{0,CL} = \frac{V_{out}}{I_{in}} = \frac{R_0}{1 + g_{mF} R_0} = \frac{1}{g_{mF}} \frac{g_{mF} R_0}{1 + g_{mF} R_0}$$



Ex.8.5

- Calculating overall transimpedance



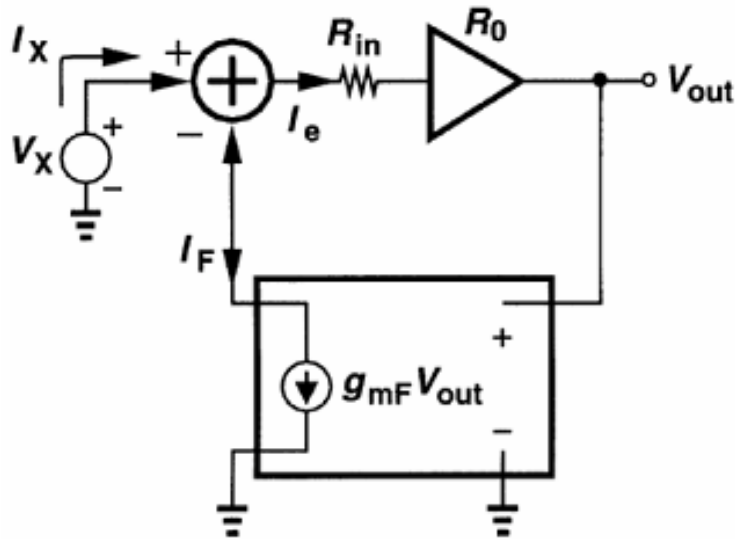
– Loop gain

$$\frac{v_f}{v_t} = -\frac{C_1}{C_1 + C_2} g_{m1} R_D$$

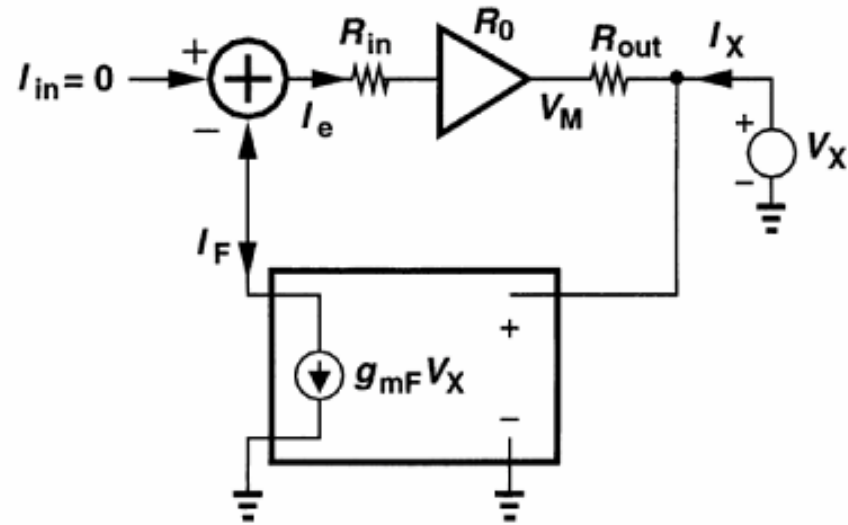
$$R_{tot} = \frac{R_D}{1 + g_{m1} R_D \cdot \frac{C_1}{C_1 + C_2}} = \frac{1}{g_{m1} \frac{C_1}{C_1 + C_2}} \frac{R_D g_{m1} \frac{C_1}{C_1 + C_2}}{1 + g_{m1} R_D \frac{C_1}{C_1 + C_2}}$$



Input & Output Resistance



$$R_{in,CL} = \frac{V_X}{I_X} = \frac{R_{in}}{1 + g_{mF} R_0}$$



$$R_{out,CL} = \frac{V_X}{I_X} = \frac{R_{out}}{1 + g_{mF} R_0}$$

- R_{out} is decreased, R_{in} is decreased

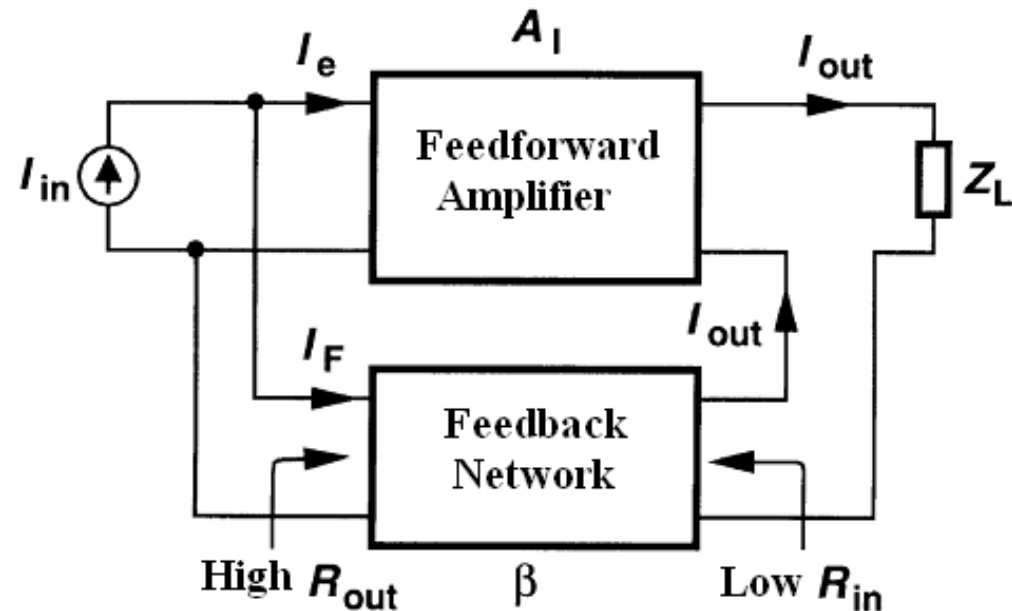


Outline

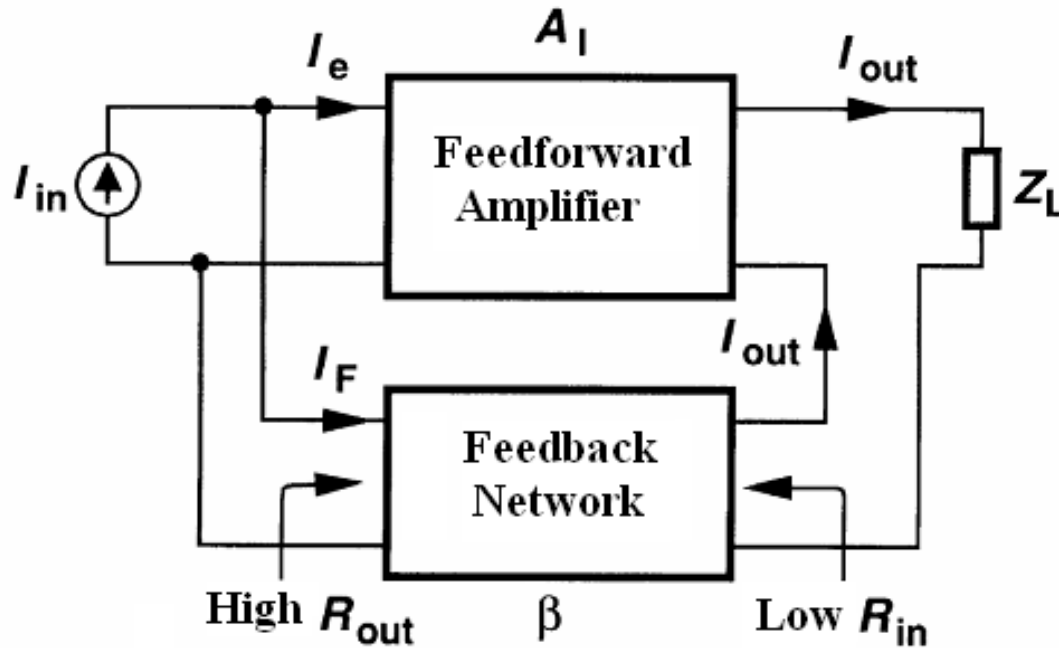
- General consideration
- Feedback topologies
 - Voltage-Voltage feedback
 - Current-Voltage feedback
 - Voltage-Current feedback
 - **Current-Current feedback**
- Effect of feedback on noise
- Feedback analysis difficulties

Current - Current Feedback

- The feedforward amplifier is an OCA
- Sense Current and return current
- Feedback network requirements:
 - Series with output
 - Parallel with input
 - Low R_{in}
 - High R_{out}



Closed-loop transfer function



$$A_{I,CL} = \frac{A_I}{1 + \beta A_I}$$

$$= \frac{1}{\beta} \frac{\beta A_I}{1 + \beta A_I}$$

$$R_{in,CL} = \frac{R_{in}}{1 + \beta A_I}, \quad R_{out,CL} = R_{out} (1 + \beta A_I)$$



Feedback Summary

Type	Sense	$R_{out,CL}$	Adding	$R_{in,CL}$	H_{CL}
VVF	V (P)	$R_{out}/(1+T)$	V (S)	$R_{in} * (1+T)$	$A_{V,OL}/(1+T)$
CVF	I (S)	$R_{out} * (1+T)$	V (S)	$R_{in} * (1+T)$	$G_{m,OL}/(1+T)$
VCF	V (P)	$R_{out}/(1+T)$	I (P)	$R_{in} / (1+T)$	$R_{0,OL}/(1+T)$
CCF	I (S)	$R_{out} * (1+T)$	I (P)	$R_{in} / (1+T)$	$A_{I,OL}/(1+T)$

T: Loop gain

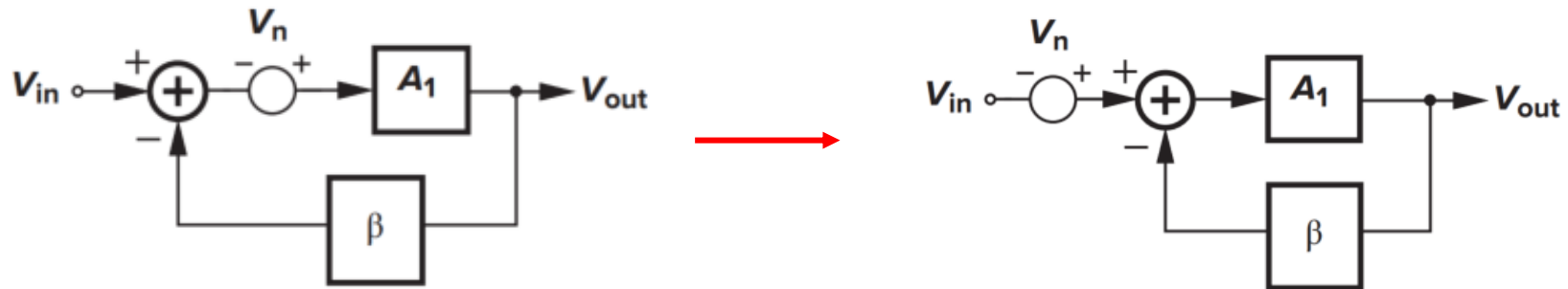
Parallel reduces resistance; **Series** increases resistance

Outline

- General consideration
 - Abstract model
 - Properties of feedback circuits
- Feedback topologies
- **Effect of feedback on noise**
- Feedback analysis difficulties
- Two-port network and loading effect

Effect of Feedback on Noise

- Feedback does not improve noise performance of circuits.



$$(V_{in} - \beta V_{out} + V_n)A_1 = V_{out}$$

$$V_{out} = (V_{in} + V_n) \frac{A_1}{1 + \beta A_1}$$

The IRN of the closed-loop circuit is the same as the open loop amp.



Outline

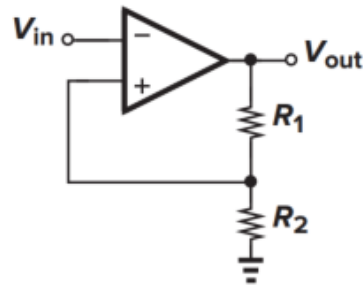
- General consideration
 - Abstract model
 - Properties of feedback circuits
- Feedback topologies
- Effect of feedback on noise
- **Feedback analysis difficulties**
- Two-port network and loading effect

Approach of Feedback Analysis

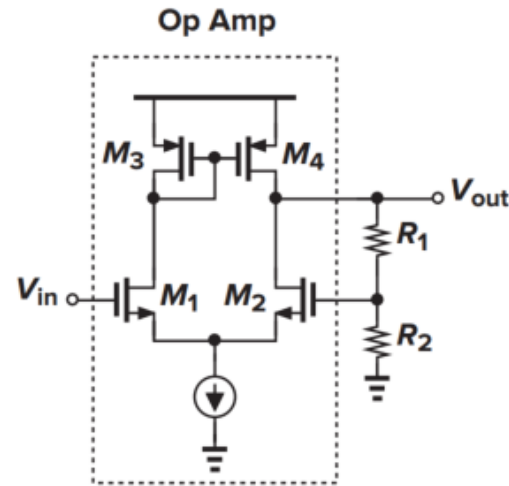
- General analysis procedure:
 - 1. Break the loop → Open-loop gain, input/output impedance
 - 2. Determine the loop gain
 - 3. Obtain the closed-loop parameters
 - 4. Using the loop gain to study stability. (Chapter 10)
- Above approach may face issues in some circuits



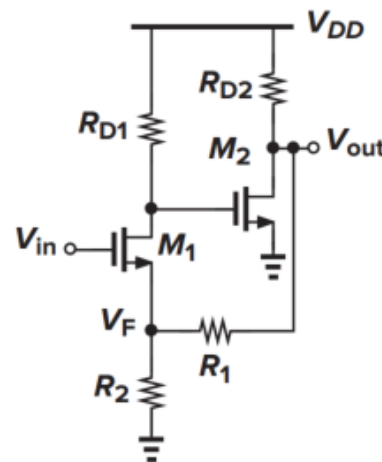
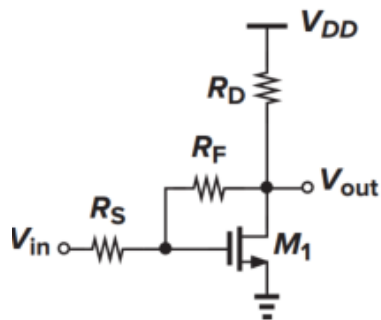
Difficulty 1: loading effect



(a)



(b)

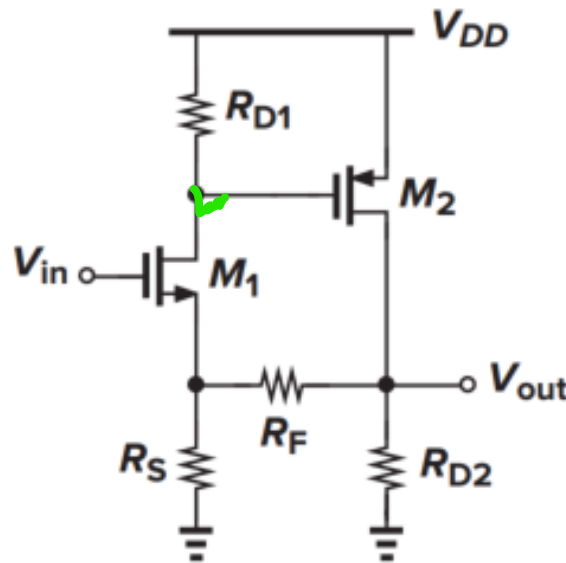


- **Output loading:**
 - Non-ideal input impedance of the feedback network
- **Input loading:**
 - Non-ideal output impedance in of the feedback network
- **How to break the loop in these cases?**



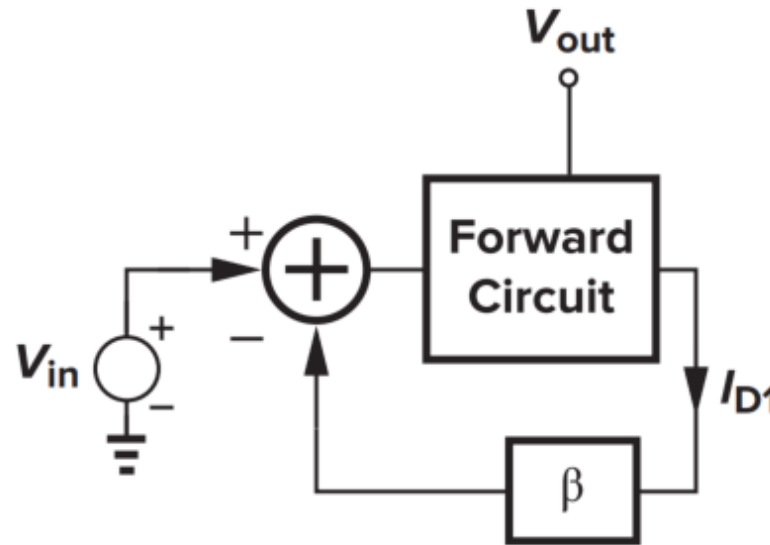
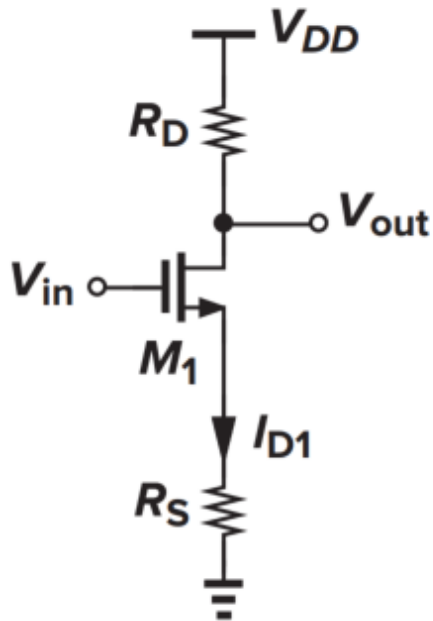
Difficulty 2: Difficult to decompose the system

- Cannot decompose the circuits into a forward amplifier and a feedback amplifier clearly
 - Does R_{D2} belong to the feedback network or the feedforward amplifier?
 - Not very clear



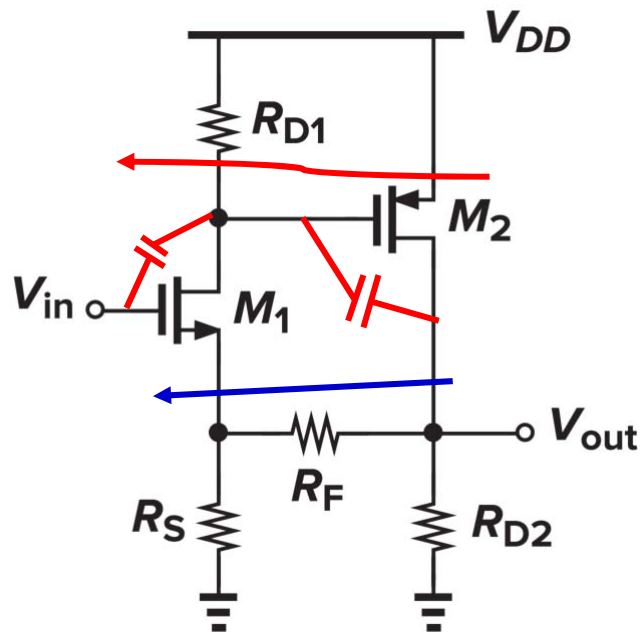
Difficulty 3: cannot be categorized clearly

- Some circuits do not readily map to the four feedback types
 - The sensed quantity is different from the output of interest (V_{out}).



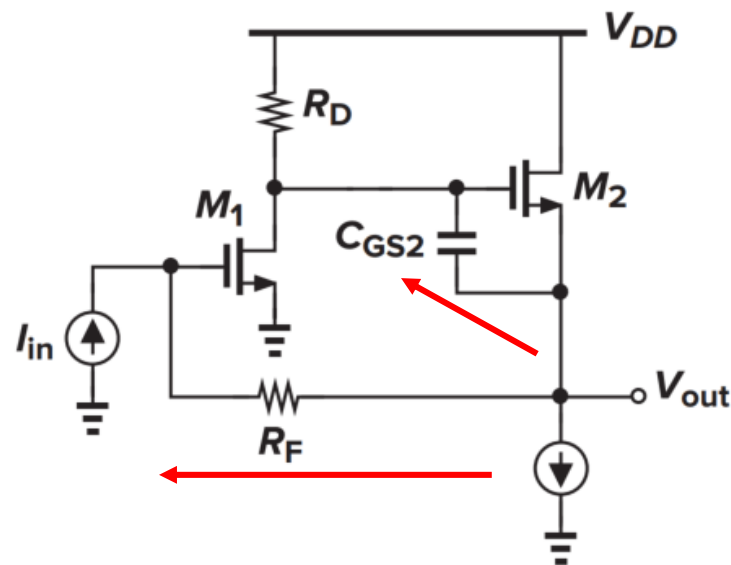
Difficulty 4: additional feedback path

- Some circuits are bilateral, allowing signals to flow from the output toward the input through another path

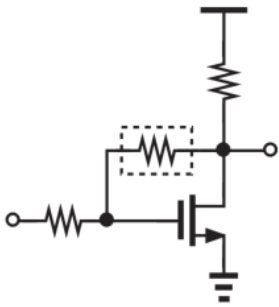
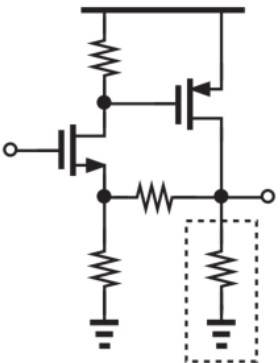
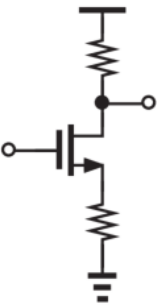
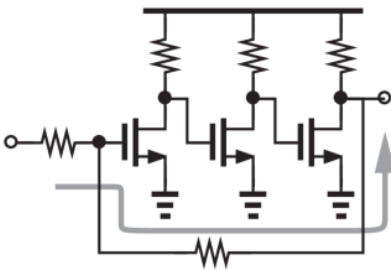
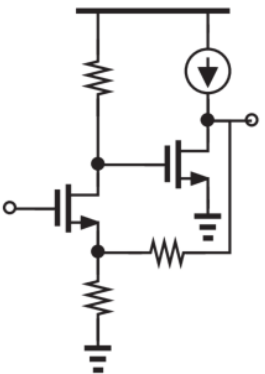


Difficulty 5: Multiple feedback mechanisms

- Some circuits contain multiple feedback loops
 - Which loop should be broken?
 - What's the exact loop gain?



Summary of the 5 difficulties

Loading	Ambiguous Decomposition	Noncanonical Topologies
		
Nonunilateral Loop	Multiple Feedback Mechanisms	
		



Three additional analysis methods

Two-Port Method	Bode's Method
• Computes open-loop and closed-loop quantities and the loop gain. • Includes loading effects. • Neglects feedforward through feedback network. • Can be applied recursively to multiple feedback mechanisms. • Does not apply to noncanonical topologies.	• Computes closed-loop quantities without breaking the loop. • Applies to any topology. • Provides loop gain only if one feedback mechanism is present.

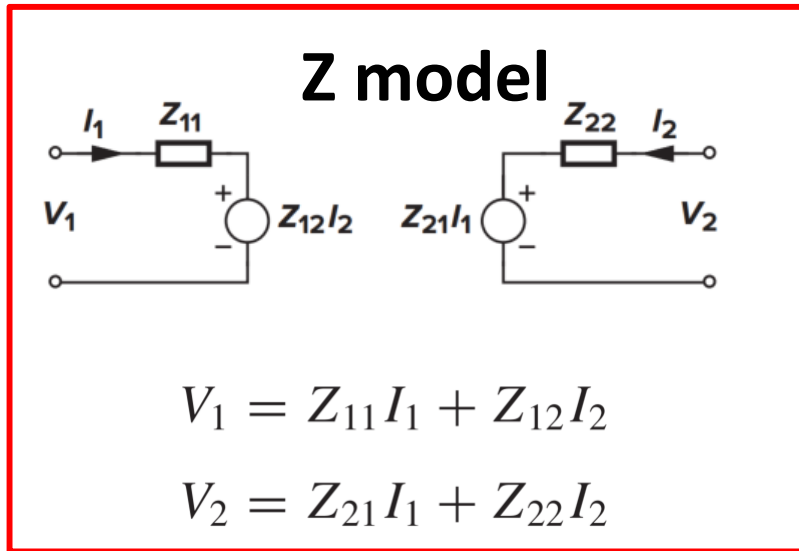
Middlebrook's Method
• Computes closed-loop quantities without breaking the loop. • Applies to any topology. • Provides loop gain only if local and global loops are distinguishable. • Reveals effect of reverse loop gain in nonunilateral loops.

Outline

- General consideration
 - Abstract model
 - Properties of feedback circuits
- Feedback topologies
- Effect of feedback on noise
- Feedback analysis difficulties
- **Two-port network and loading effect**

Two-Port Network Models

- Many circuits' **small-signal model** can be modeled as a 2-port linear network



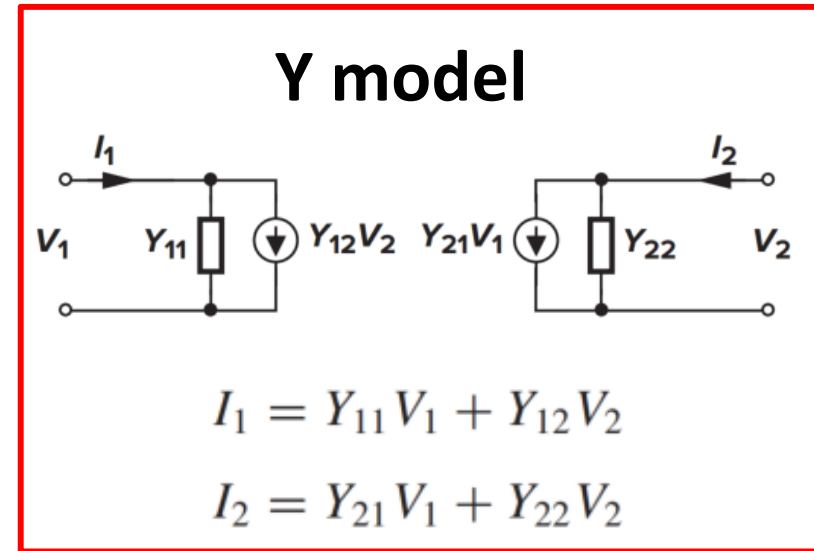
Input/output impedance

$$Z_{11} = V_1 / I_1 \Big|_{I_2=0}$$

$$Z_{22} = V_2 / I_2 \Big|_{I_1=0}$$

$$Z_{11} \stackrel{?}{=} 1/Y_{11}$$

Why?



Input/output admittance

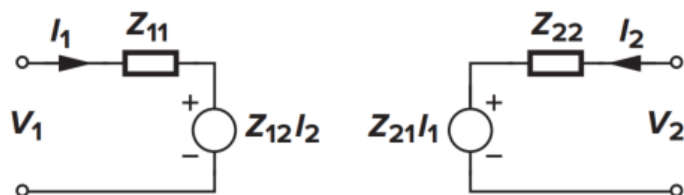
$$Y_{11} = I_1 / V_1 \Big|_{V_2=0}$$

$$Y_{22} = I_2 / V_2 \Big|_{V_1=0}$$

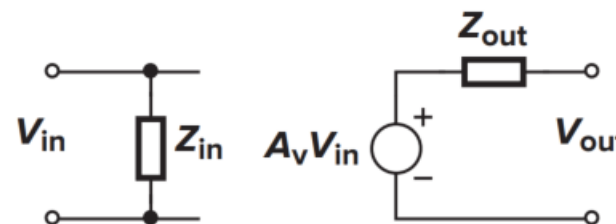


Amplifier model **vs.** Two-Port Network Model

Z model



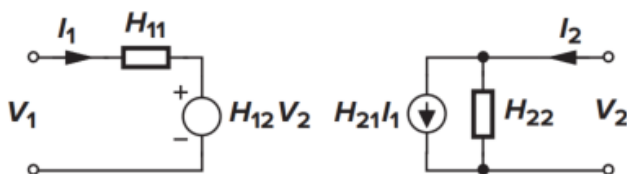
Amplifier model



- $Z_{12}I_2$ represents the internal feedback
- If $I_2 Z_{12} = 0$ (output open, or no internal feedback) $\rightarrow Z_{11} = Z_{in}$
- Z_{22} is obtained with input open while Z_{out} is obtained with input short, they may not equal
- Output generator $Z_{21}I_1$ is controlled by I_1 instead of V_1 , which is not suitable for our MOS amplifiers

H model and G model

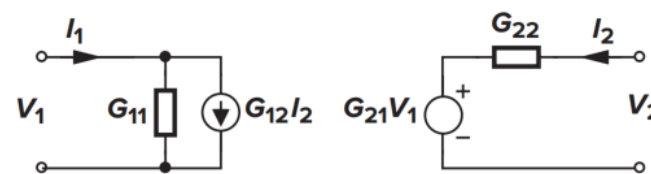
H model



$$V_1 = H_{11} I_1 + H_{12} V_2$$

$$I_2 = H_{21} I_1 + H_{22} V_2$$

G model



$$I_1 = G_{11} V_1 + G_{12} I_2$$

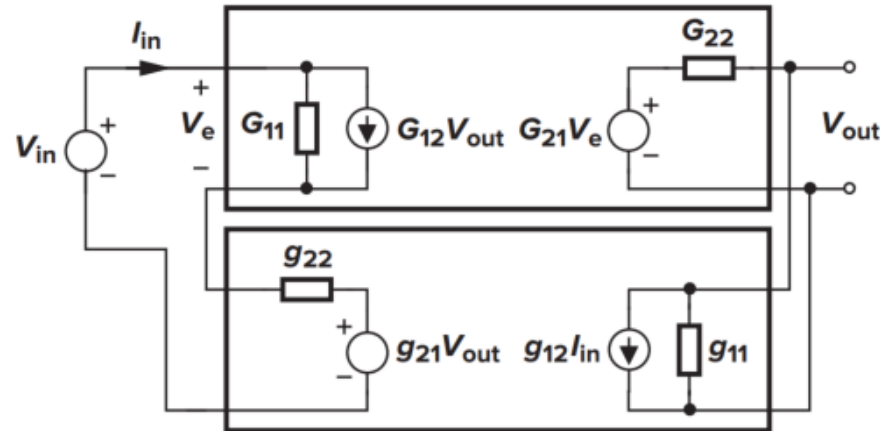
$$V_2 = G_{21} V_1 + G_{22} I_2$$

- **G model agrees with our voltage amplifier model**
 - $G_{12} I_2 = 0 \rightarrow G_{11} = I_1 / V_1$
 - $V_1 = 0$ (short), $\rightarrow G_{22} = V_2 / I_2$

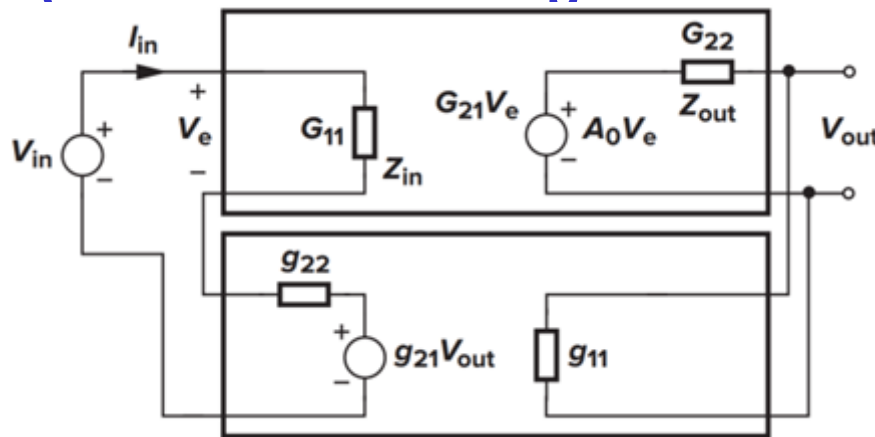


Loading in a VVF circuit

- Use G model to analyze the circuit



- Simplified G model: $G_{12} = 0$ (internal feedback), $g_{12} = 0$ (feedforward through the feedback network)

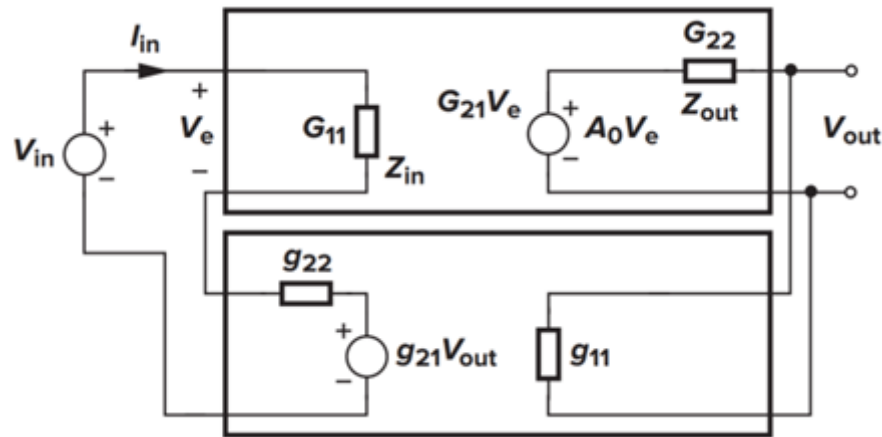


$$G_{11} = Z_{in}, G_{21} = A_0, G_{22} = Z_{out}$$

$$V_{in} = V_e + g_{22} \frac{V_e}{Z_{in}} + g_{21} V_{out}$$

$$g_{11} V_{out} + \frac{V_{out} - A_0 V_e}{Z_{out}} = 0$$





$$V_{in} = V_e + g_{22} \frac{V_e}{Z_{in}} + g_{21} V_{out}$$

$$g_{11} V_{out} + \frac{V_{out} - A_0 V_e}{Z_{out}} = 0$$



$$\frac{V_{out}}{V_{in}} = \frac{A_0}{(1 + \frac{g_{22}}{Z_{in}})(1 + g_{11} Z_{out}) + g_{21} A_0}$$

Written in a standard form: $A_{v,CL} = A_{v,open} / (1 + \beta A_{v,open})$

$$\frac{V_{out}}{V_{in}} = \frac{\frac{A_0}{(1 + \frac{g_{22}}{Z_{in}})(1 + g_{11} Z_{out})}}{1 + g_{21} \frac{A_0}{(1 + \frac{g_{22}}{Z_{in}})(1 + g_{11} Z_{out})}}$$



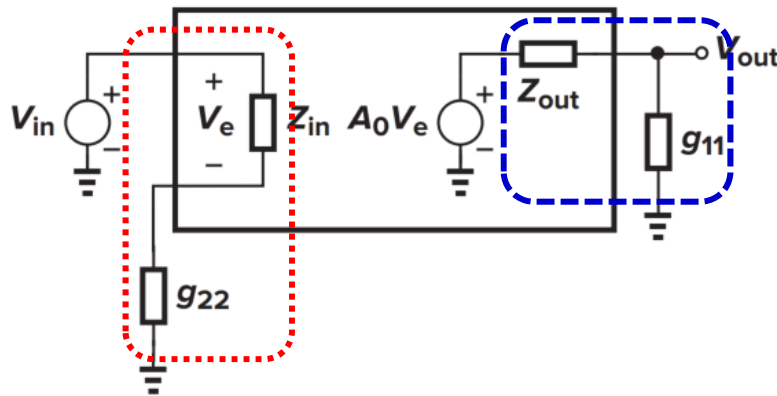
$$A_{v,open} = \frac{A_0}{(1 + \frac{g_{22}}{Z_{in}})(1 + g_{11} Z_{out})}$$

$$\beta = g_{21}$$



Open-loop gain is affected by the loading effect

$$A_{v,open} = \frac{A_0}{\left(1 + \frac{g_{22}}{Z_{in}}\right) (1 + g_{11} Z_{out})}$$



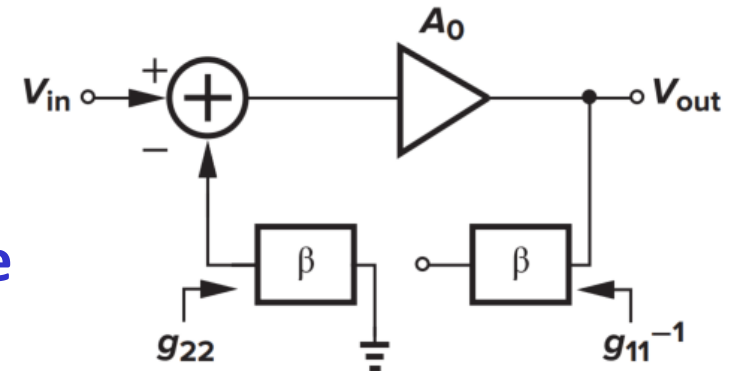
- Non-ideal input/output impedance of the feedback network

→ the **open-loop gain** is attenuated by 2 voltage dividers

Calculating g_{22} and g_{11}

$$g_{11} = \frac{I_1}{V_1} \Big|_{I_2=0}$$

$$g_{22} = \frac{V_2}{I_2} \Big|_{V_1=0}$$



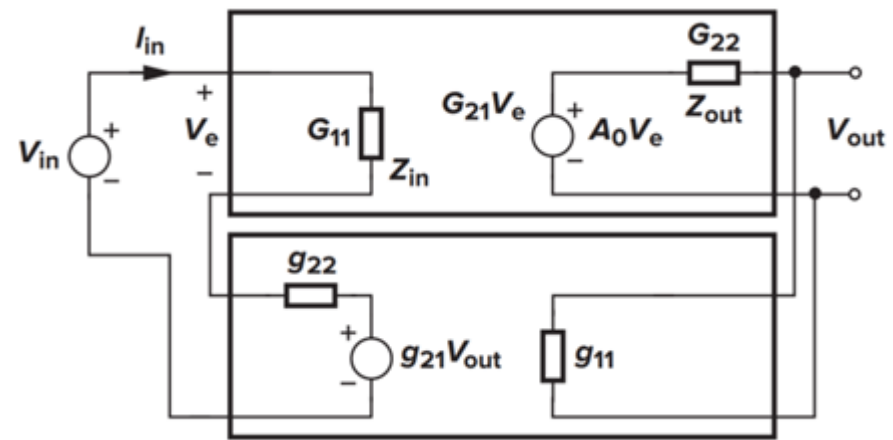
Application of the method

- Calculate the loop gain

$$\text{Loop Gain} = \beta A_{v,\text{open}} = g_{21} \frac{A_0}{(1 + g_{22} / Z_{\text{in}})(1 + g_{11} Z_{\text{out}})}$$

$$Z_{\text{in},\text{CL}} = (Z_{\text{in}} + g_{22})(1 + \beta A_{v,\text{open}})$$

$$Z_{\text{out},\text{CL}} = \frac{(Z_{\text{out}} \parallel \frac{1}{g_{11}})}{(1 + \beta A_{v,\text{open}})}$$

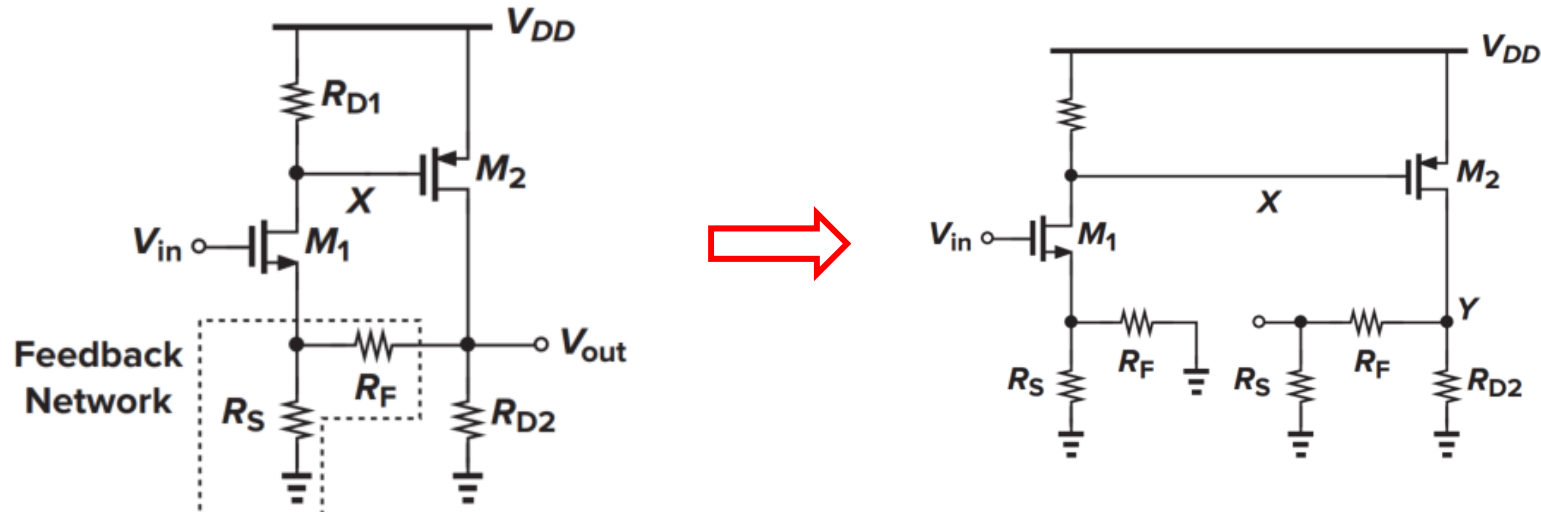


- Note that the results are obtained with $G_{12}=0$ and $g_{12} = 0$.
 - Maybe more complex when $G_{12} \neq 0$ and $g_{12} \neq 0$.



Ex. 8.12

- Calculate $A_{v,open}$ and $A_{v,CL}$, assuming $\lambda = \gamma = 0$



$$A_{v,open} = \frac{V_Y}{V_{in}} = \frac{-R_{D1}}{R_F \parallel R_S + 1/g_{m1}} \{-g_{m2}[R_{D2} \parallel (R_F + R_S)]\}$$

- $g_{21} = V_2/V_1$ with $I_2=0$

$$g_{21} = R_S/(R_F + R_S)$$

$$\Rightarrow A_{v,closed} = A_{v,open}/(1 + g_{21}A_{v,open})$$



Extending study by yourself

- Loading effect in other feedback topologies
- 8.6 Bode's Analysis of feedback
- 8.7 Middlebrook's Method
- 8.8 Loop gain calculation issues
- 8.9 Alternative analysis of Bode's method

Summary

- General consideration
 - Abstract model
 - Properties of feedback circuits
- Feedback topologies
- Effect of feedback on noise
- Feedback analysis difficulties
- Two-port network and loading effect